

BFS Traversal

Step 1: Start at E

Legend

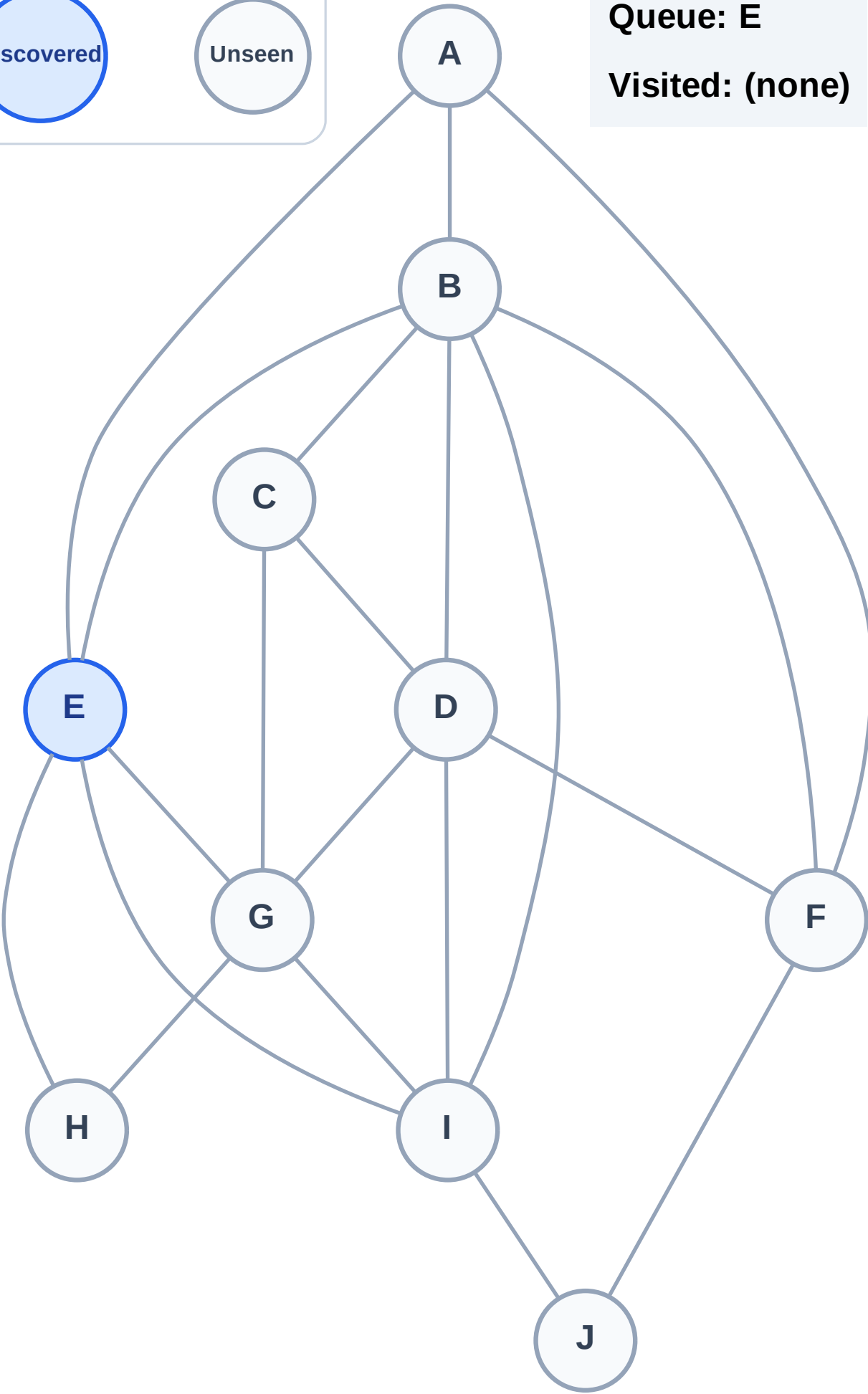
Complete

Processing

Discovered

Unseen

Queue: E
Visited: (none)



BFS Traversal

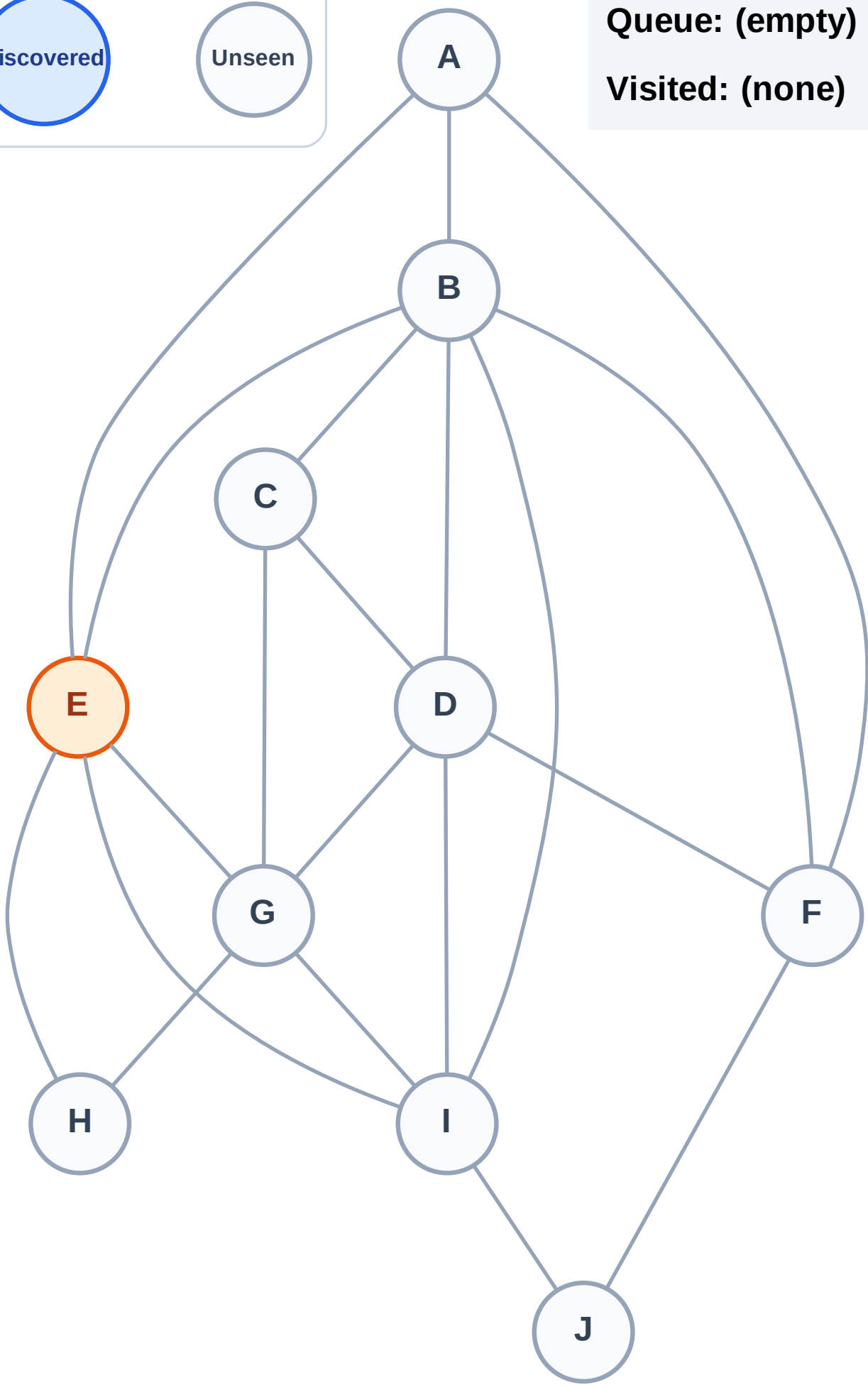
Step 2: Process node E

Legend



Queue: (empty)

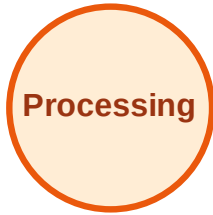
Visited: (none)



BFS Traversal

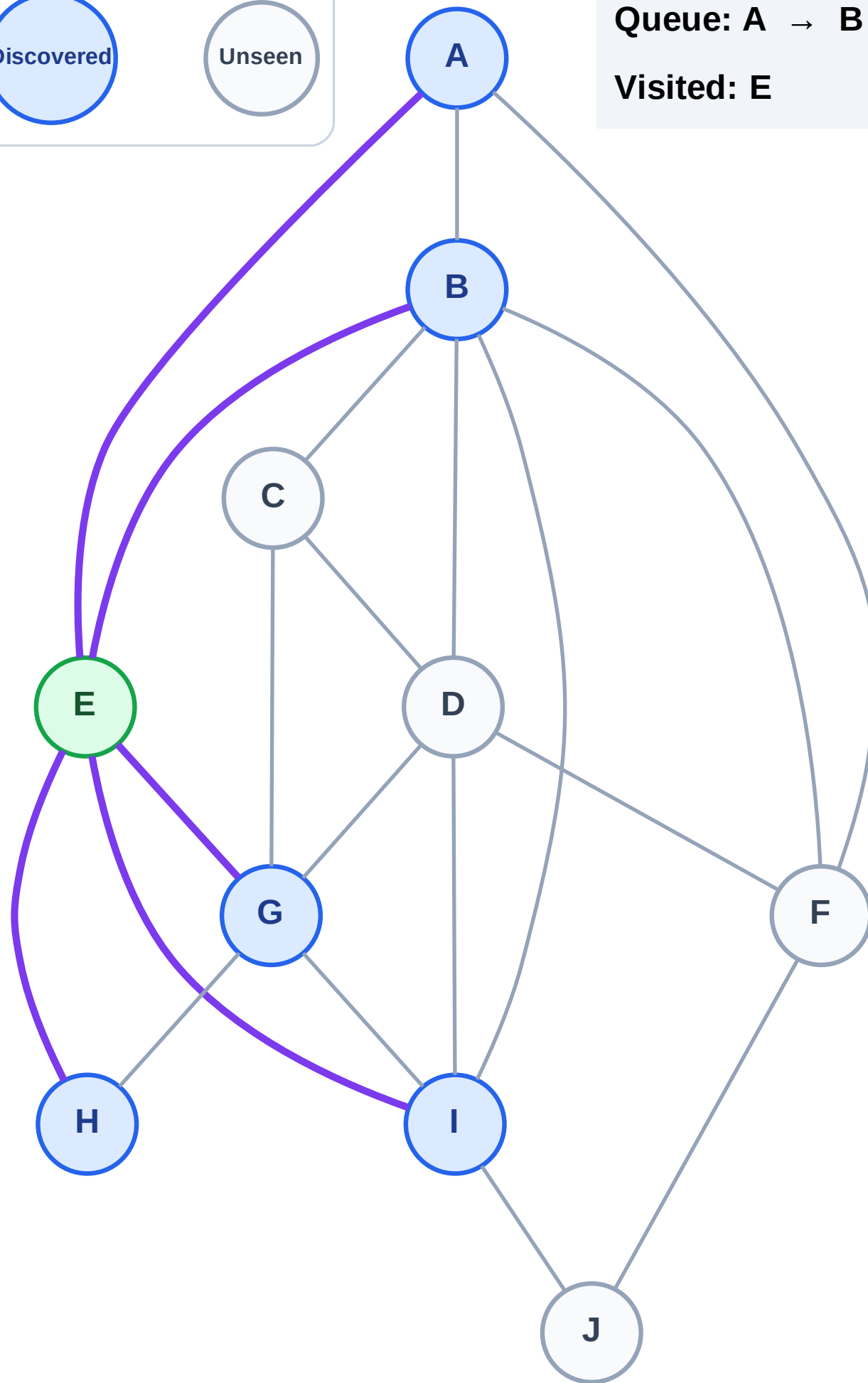
Step 3: Discover A, B, G, H, I from E

Legend



Queue: A → B → G → H → I

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

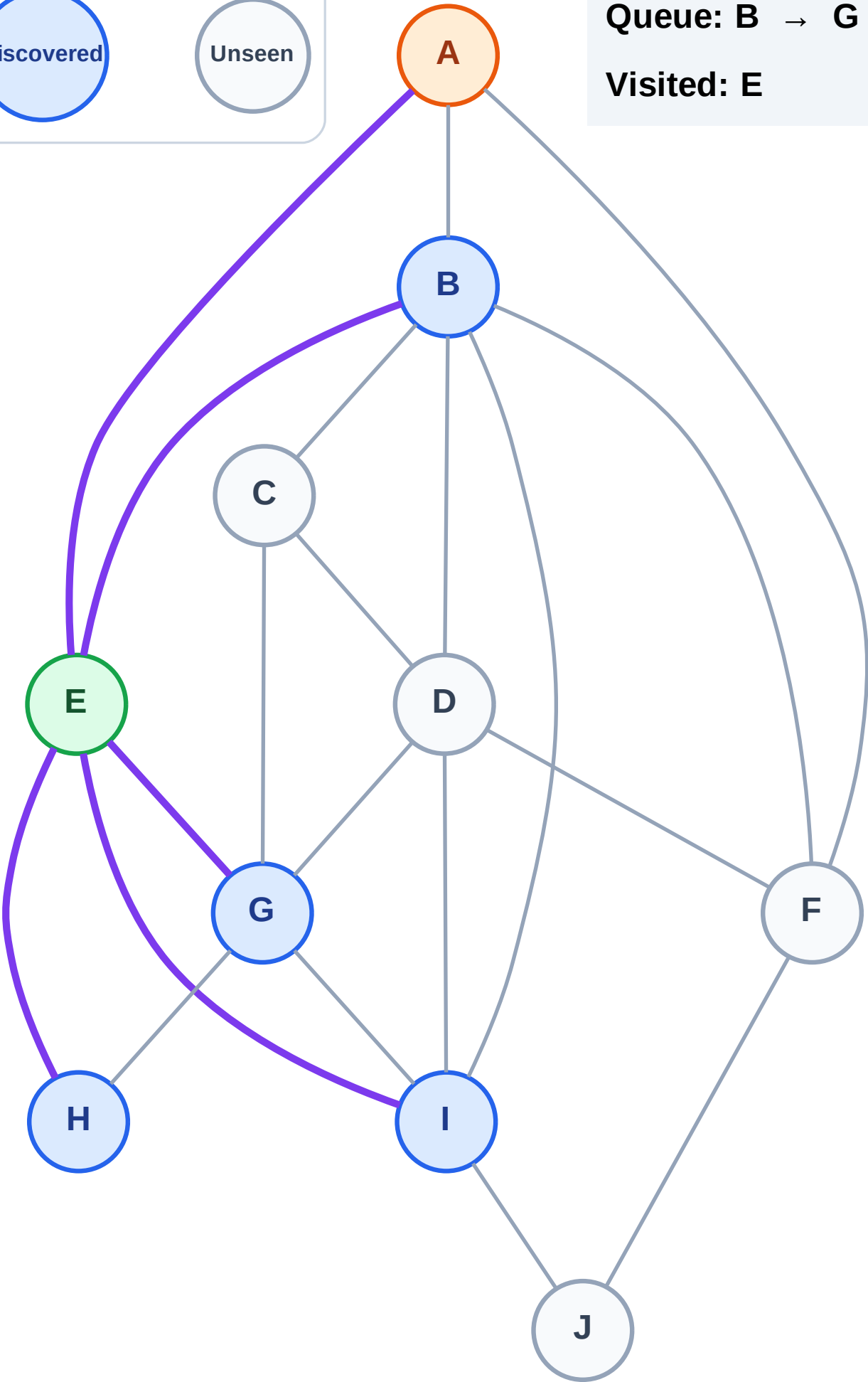
Processing

Discovered

Unseen

Queue: B → G → H → I

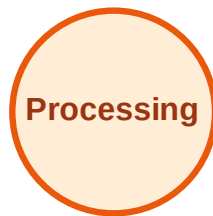
Visited: E



BFS Traversal

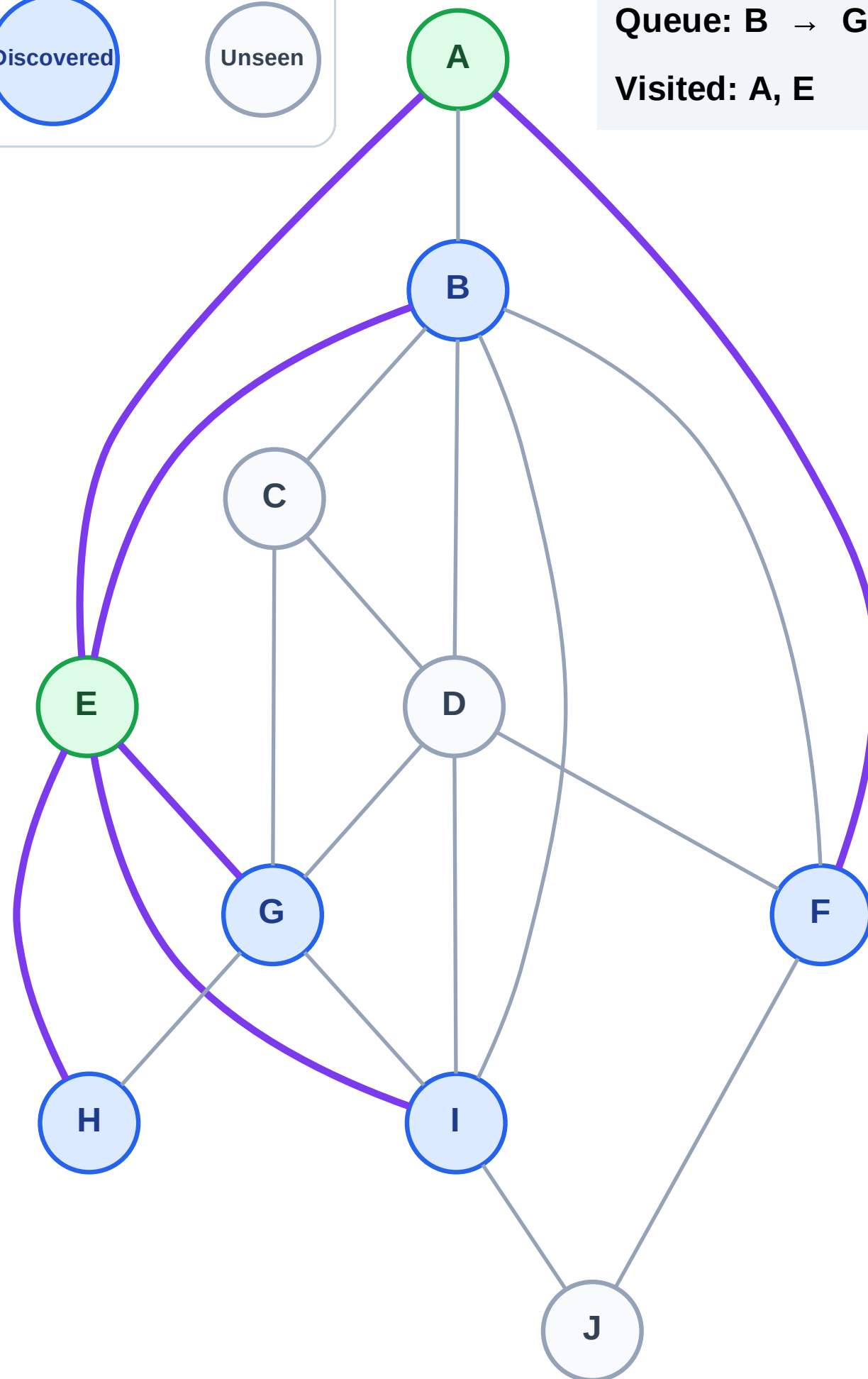
Step 5: Discover F from A

Legend



Queue: B → G → H → I → F

Visited: A, E



BFS Traversal

Step 6: Process node B

Legend

Complete

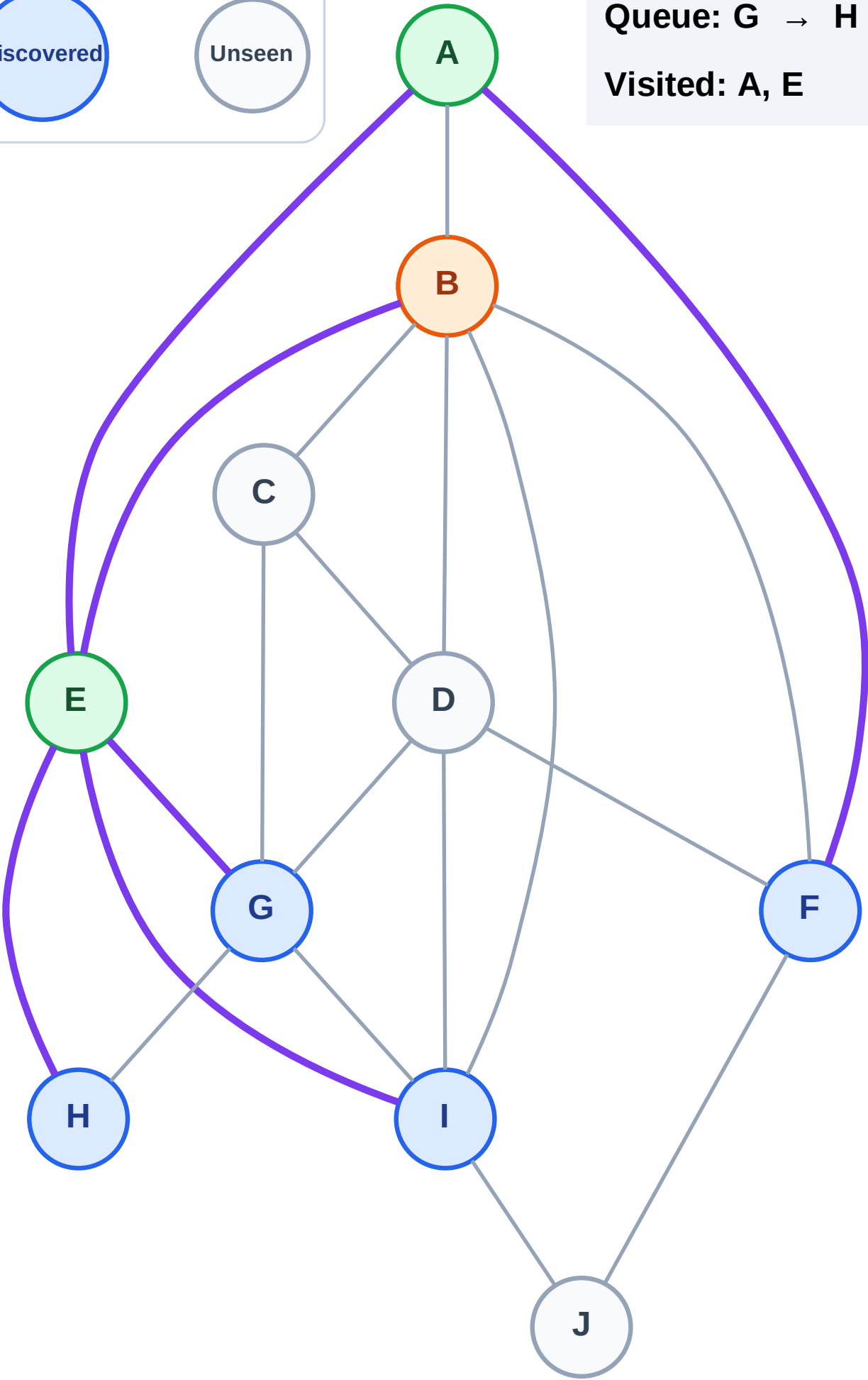
Processing

Discovered

Unseen

Queue: G → H → I → F

Visited: A, E



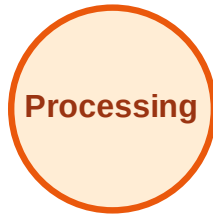
BFS Traversal

Step 7: Discover C, D from B

Legend



Complete



Processing



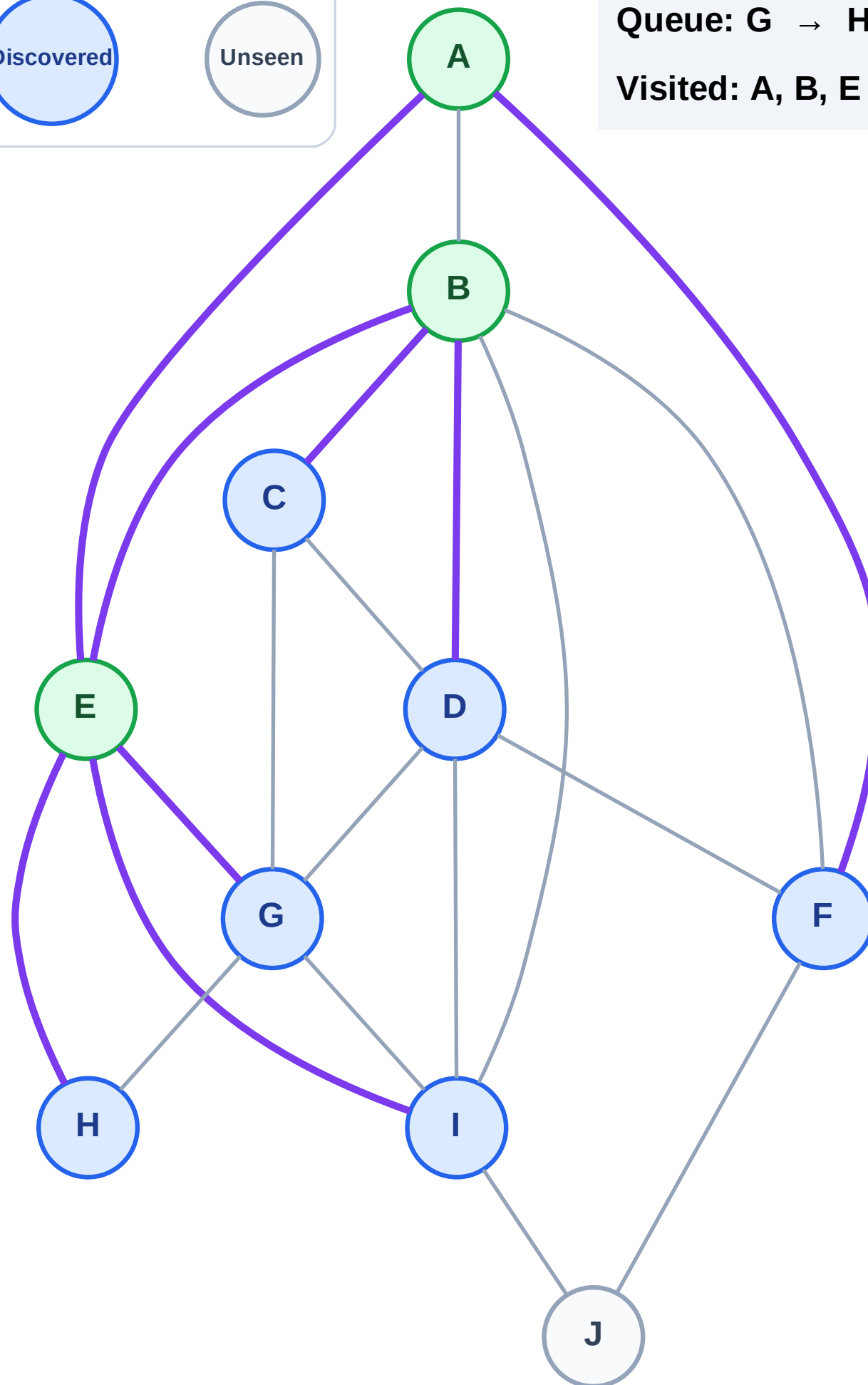
Discovered



Unseen

Queue: G → H → I → F → C → D

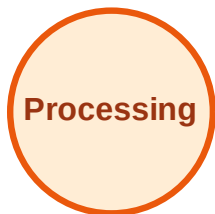
Visited: A, B, E



BFS Traversal

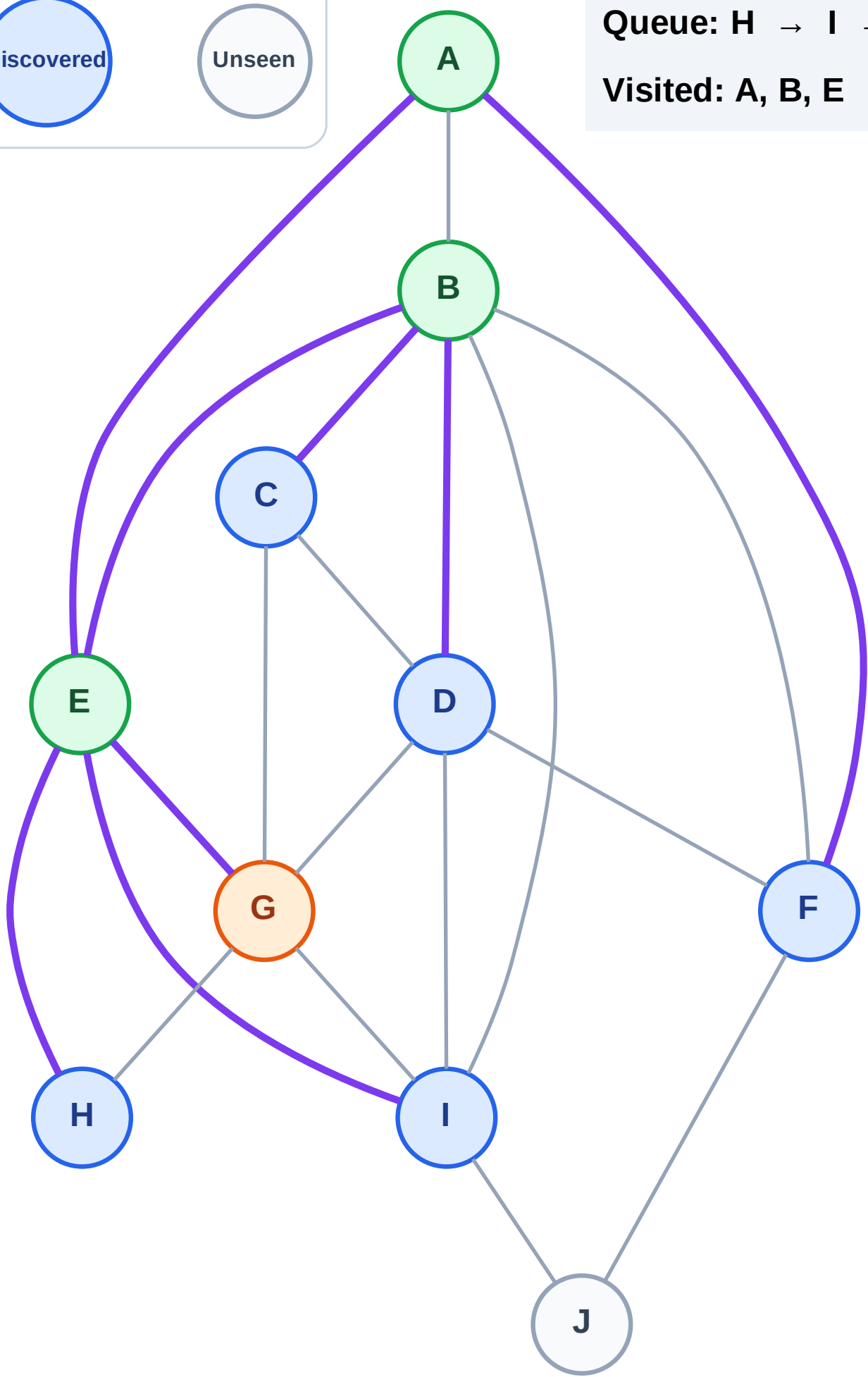
Step 8: Process node G

Legend



Queue: H → I → F → C → D

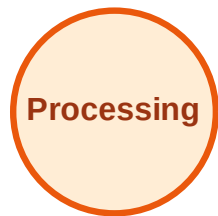
Visited: A, B, E



BFS Traversal

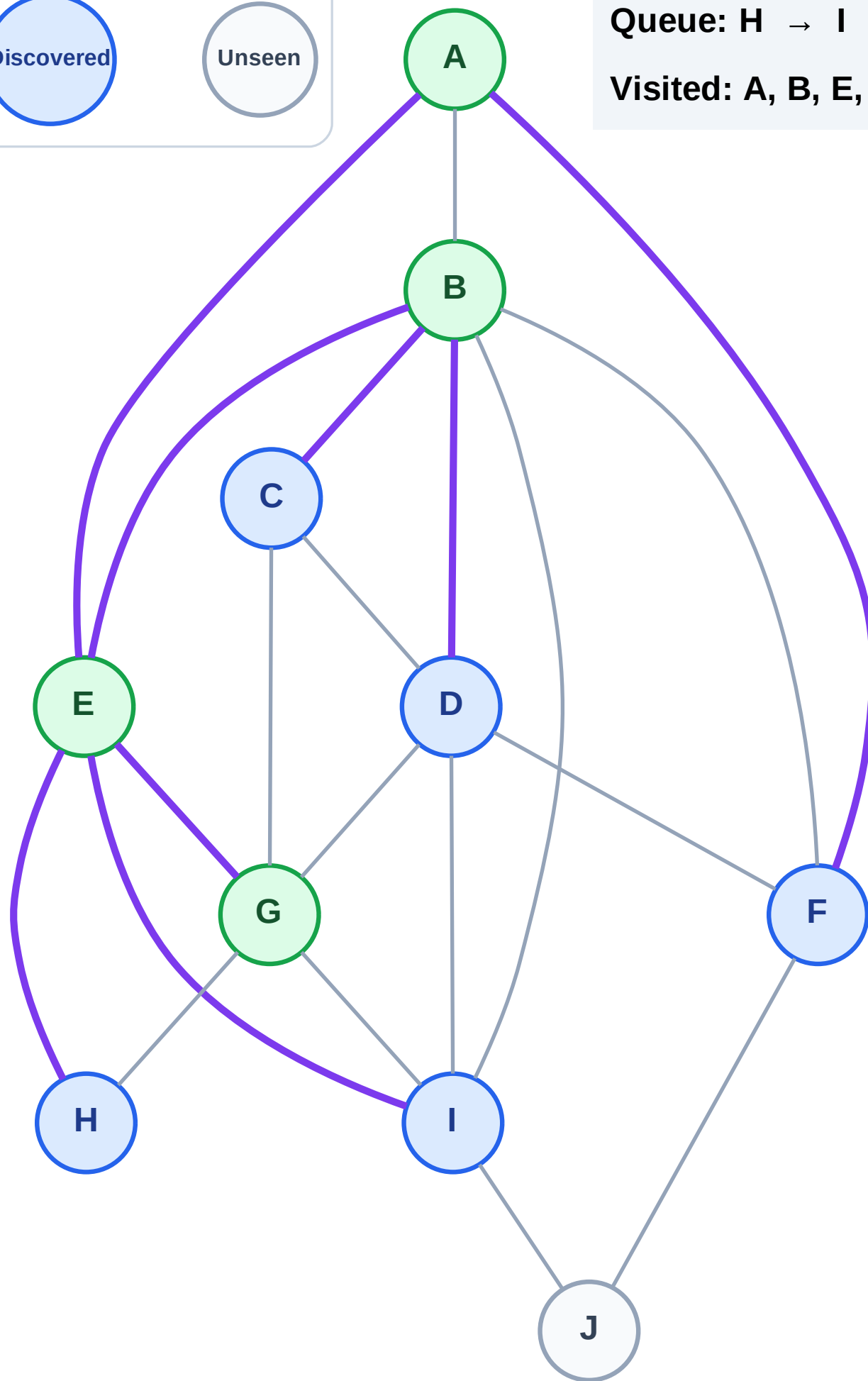
Step 9: No new neighbors from G

Legend



Queue: H → I → F → C → D

Visited: A, B, E, G



BFS Traversal

Step 10: Process node H

Legend

Complete

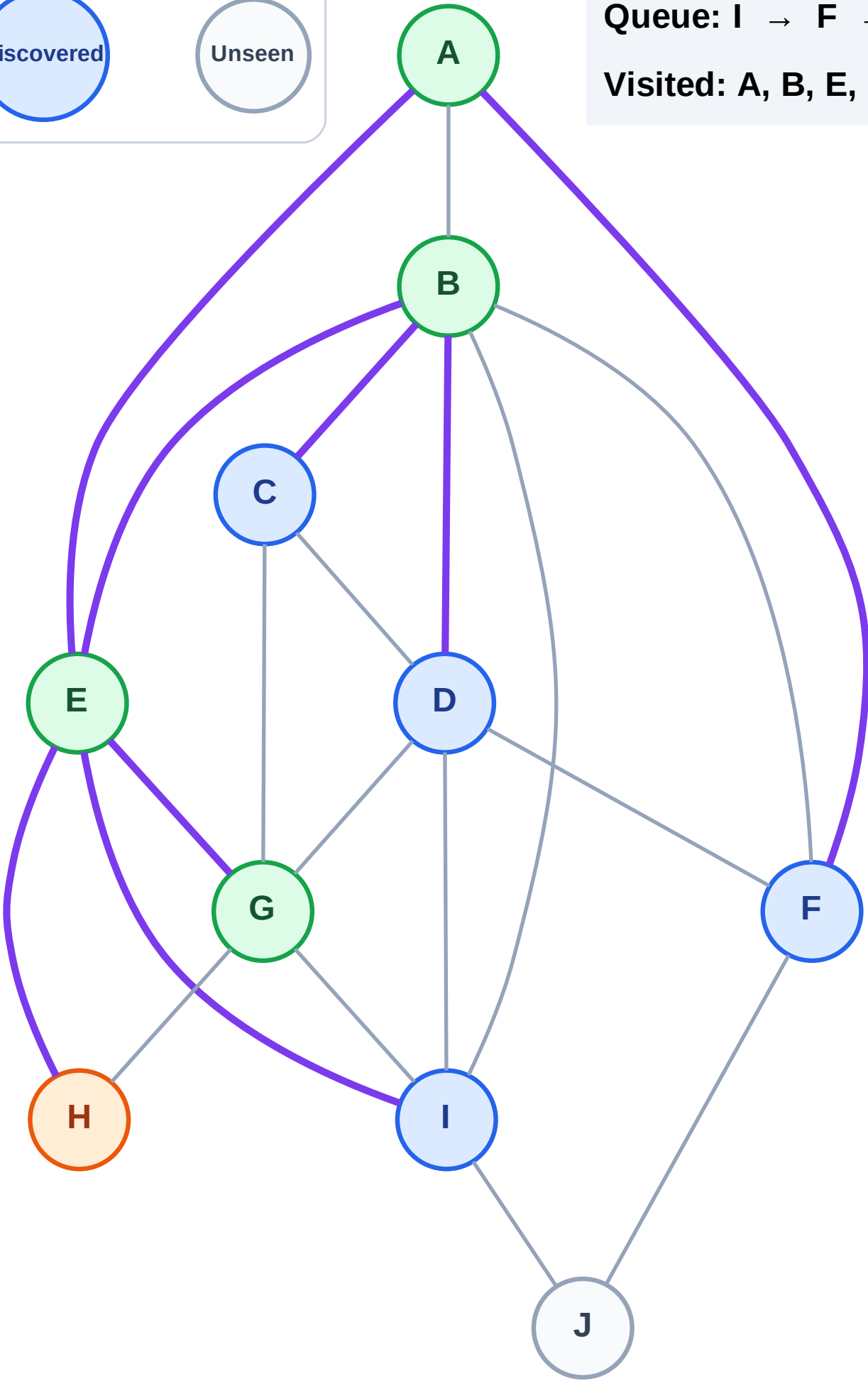
Processing

Discovered

Unseen

Queue: I → F → C → D

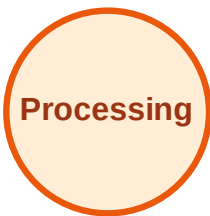
Visited: A, B, E, G



BFS Traversal

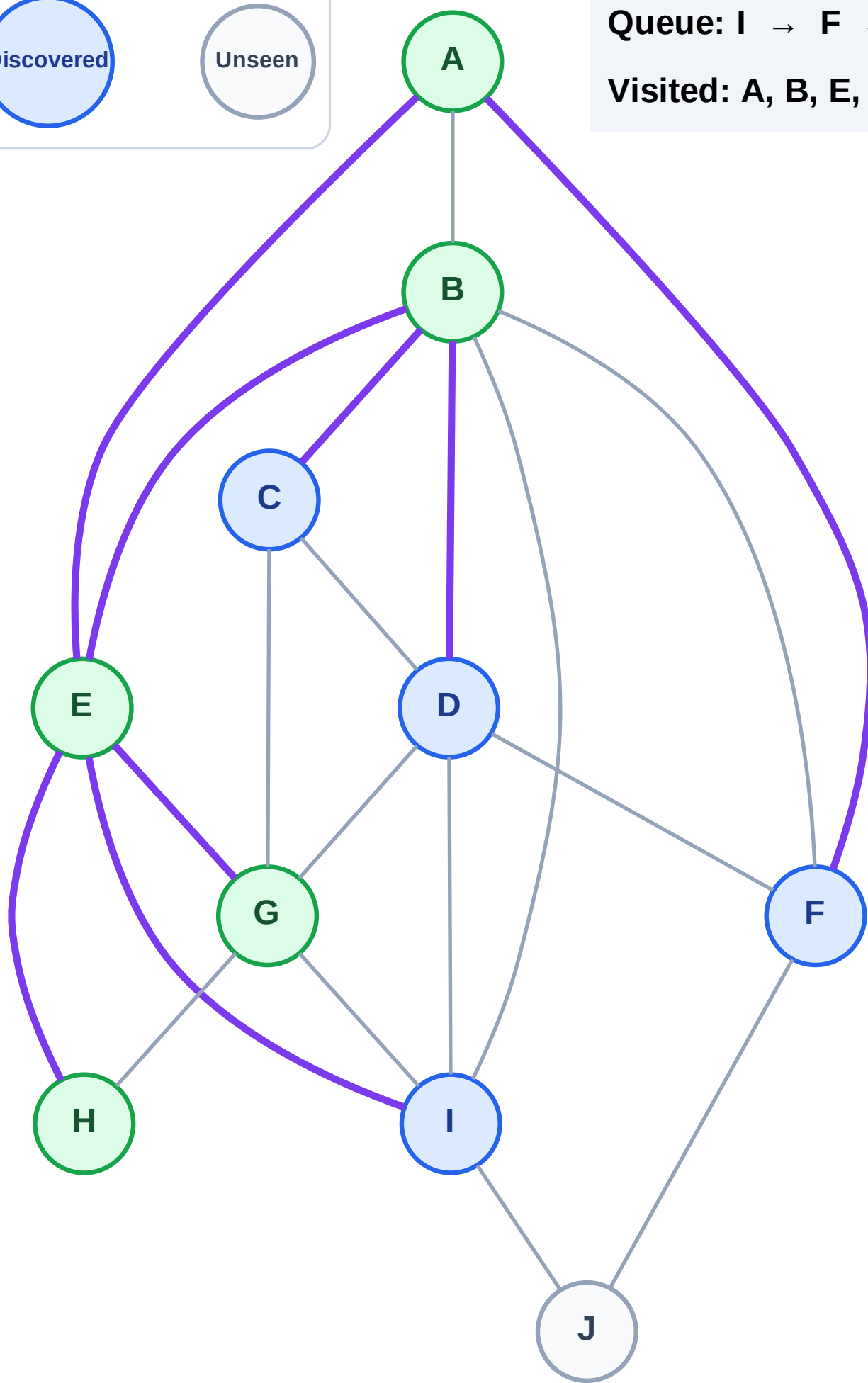
Step 11: No new neighbors from H

Legend



Queue: I → F → C → D

Visited: A, B, E, G, H



BFS Traversal

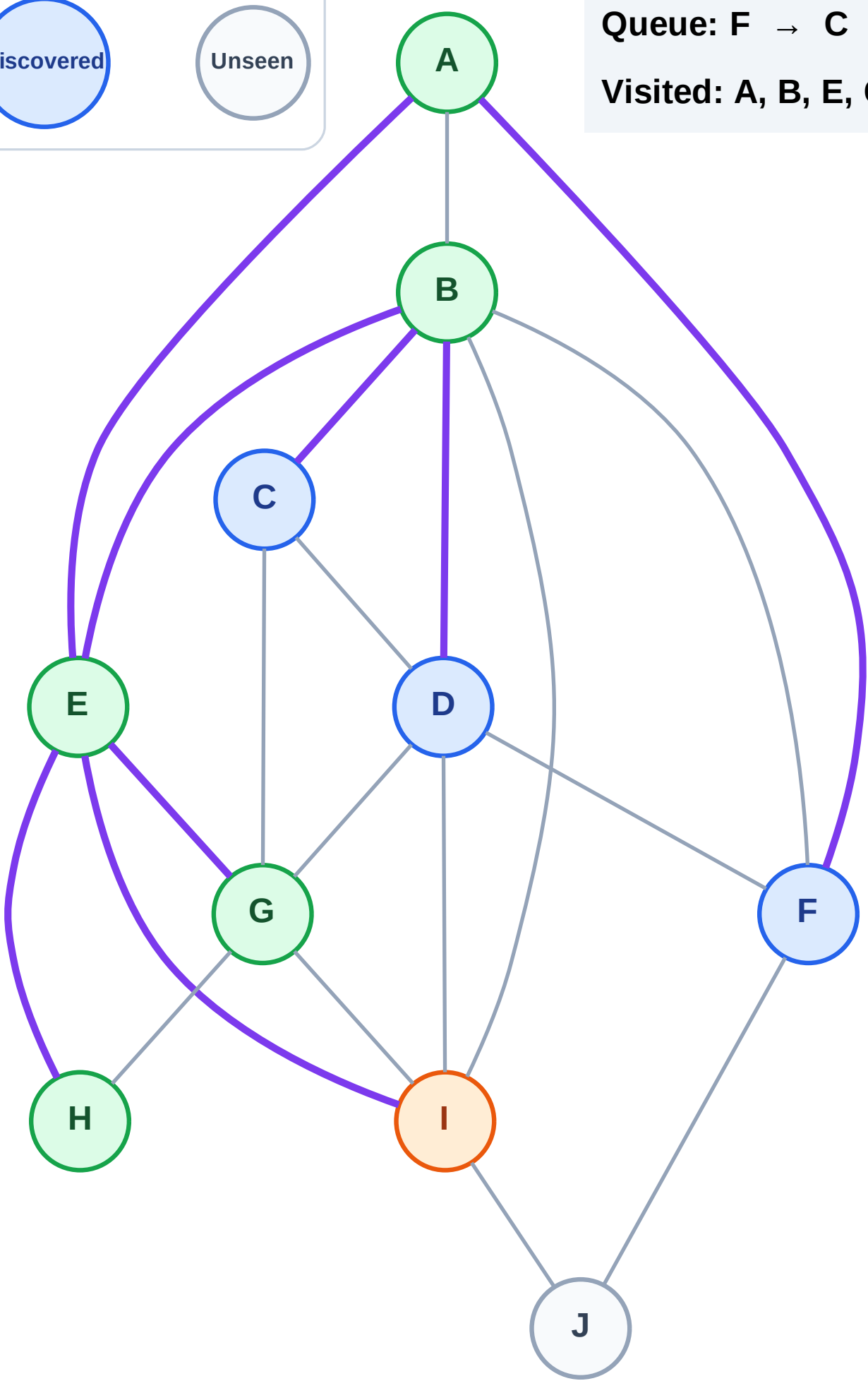
Step 12: Process node I

Legend



Queue: F → C → D

Visited: A, B, E, G, H



BFS Traversal

Step 13: Discover J from I

Legend

Complete

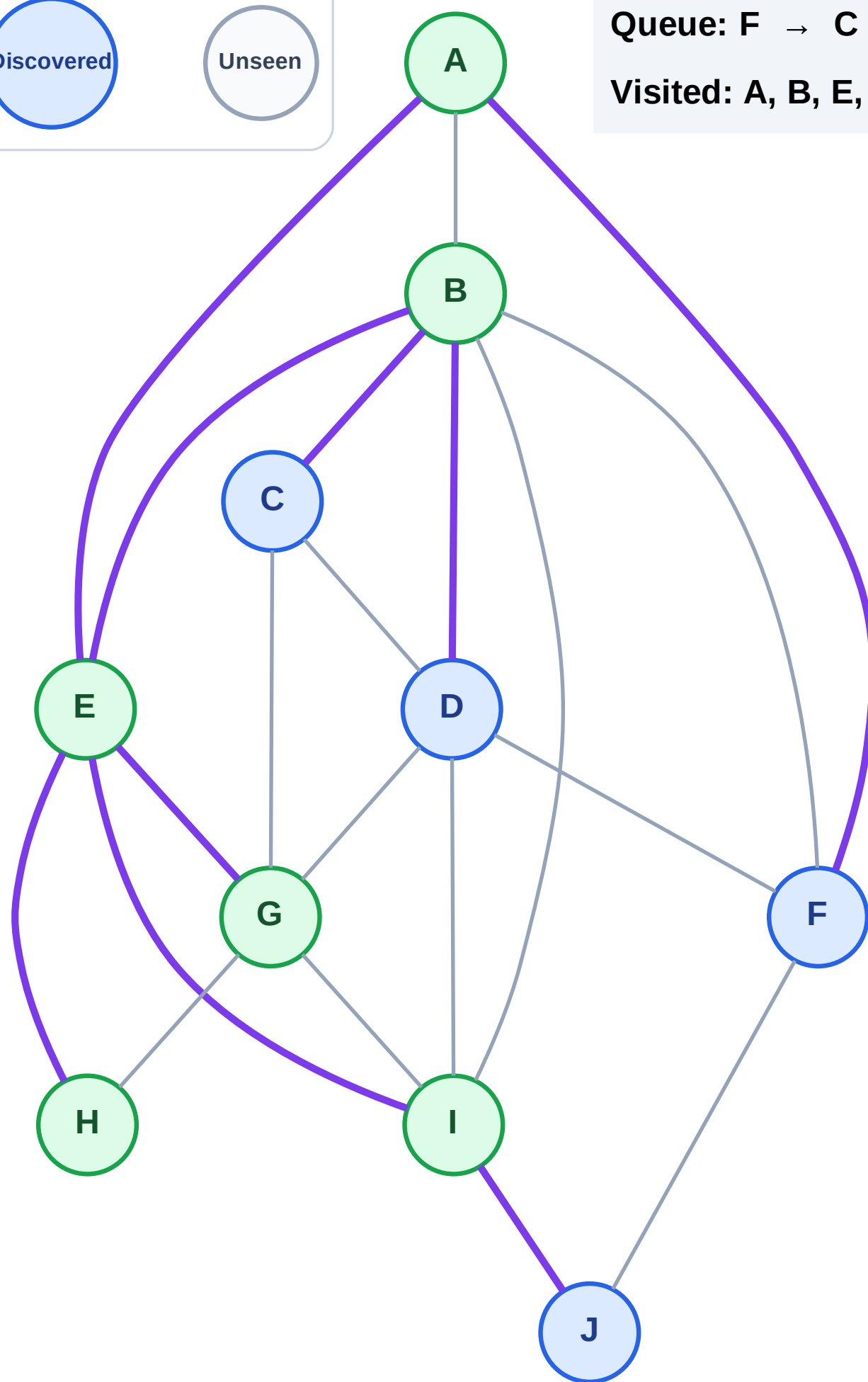
Processing

Discovered

Unseen

Queue: F → C → D → J

Visited: A, B, E, G, H, I



BFS Traversal

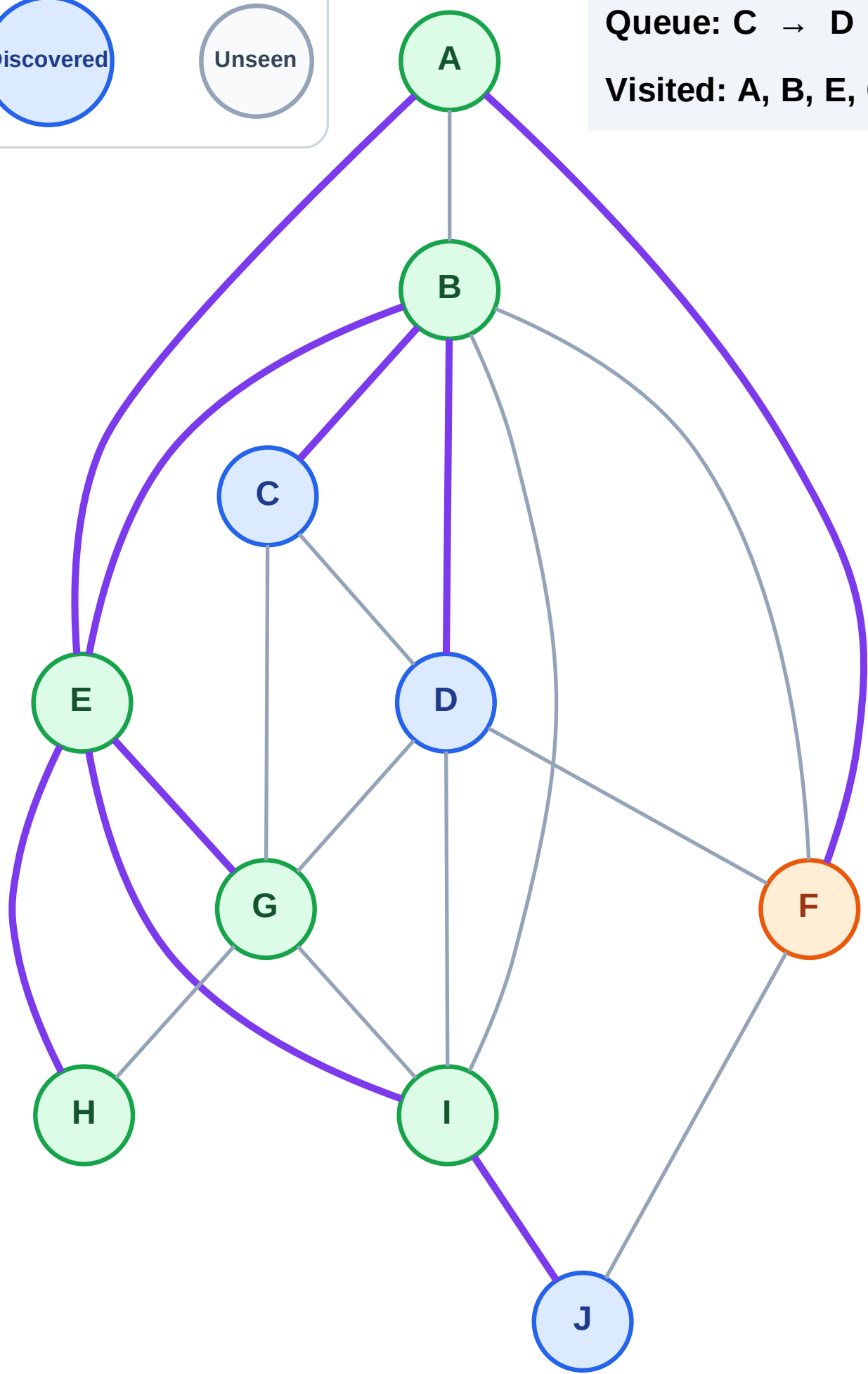
Step 14: Process node F

Legend



Queue: C → D → J

Visited: A, B, E, G, H, I



BFS Traversal

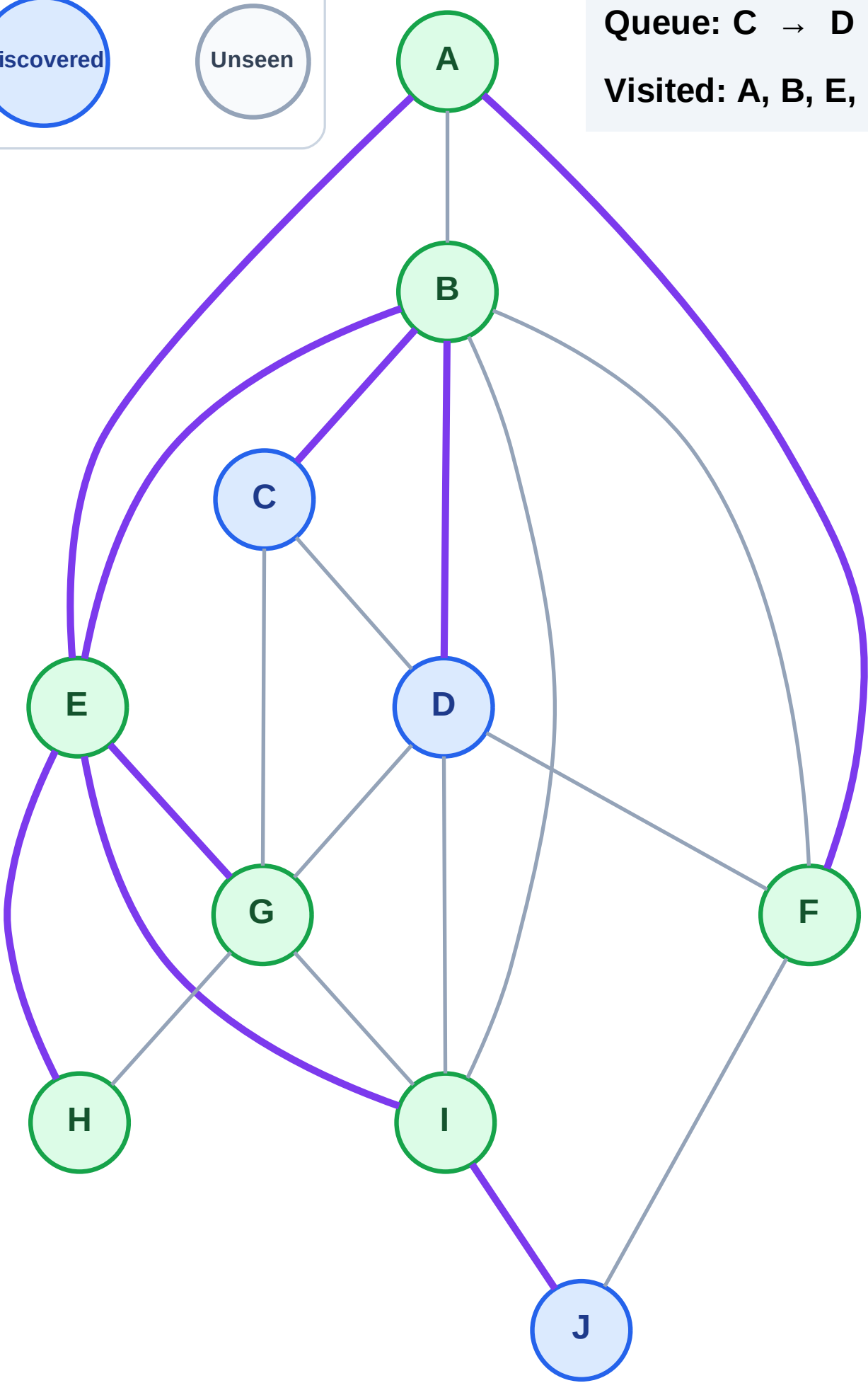
Step 15: No new neighbors from F

Legend



Queue: C → D → J

Visited: A, B, E, F, G, H, I



BFS Traversal

Step 16: Process node C

Legend

Complete

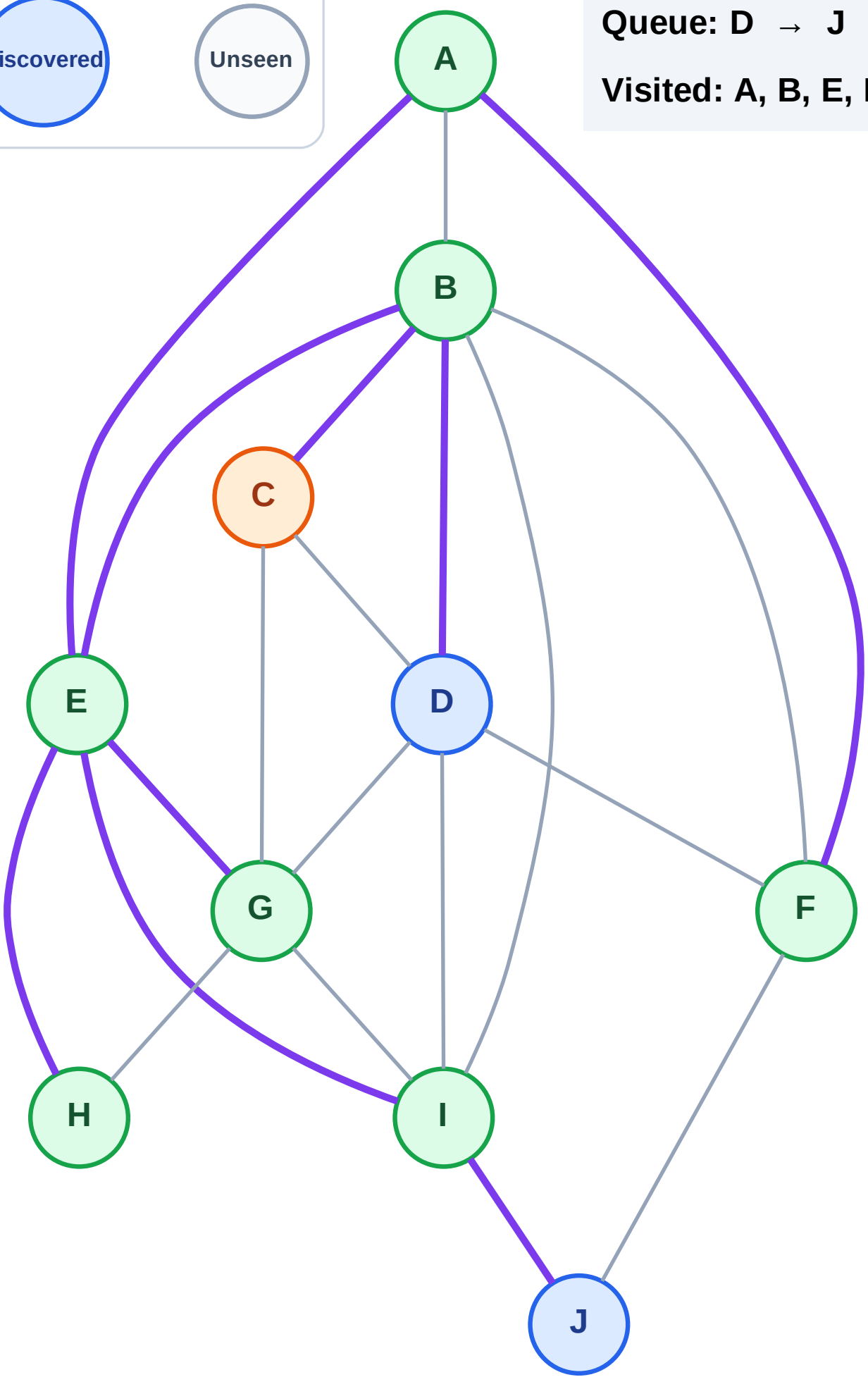
Processing

Discovered

Unseen

Queue: D → J

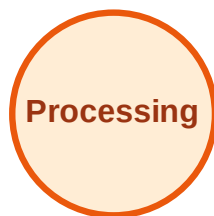
Visited: A, B, E, F, G, H, I



BFS Traversal

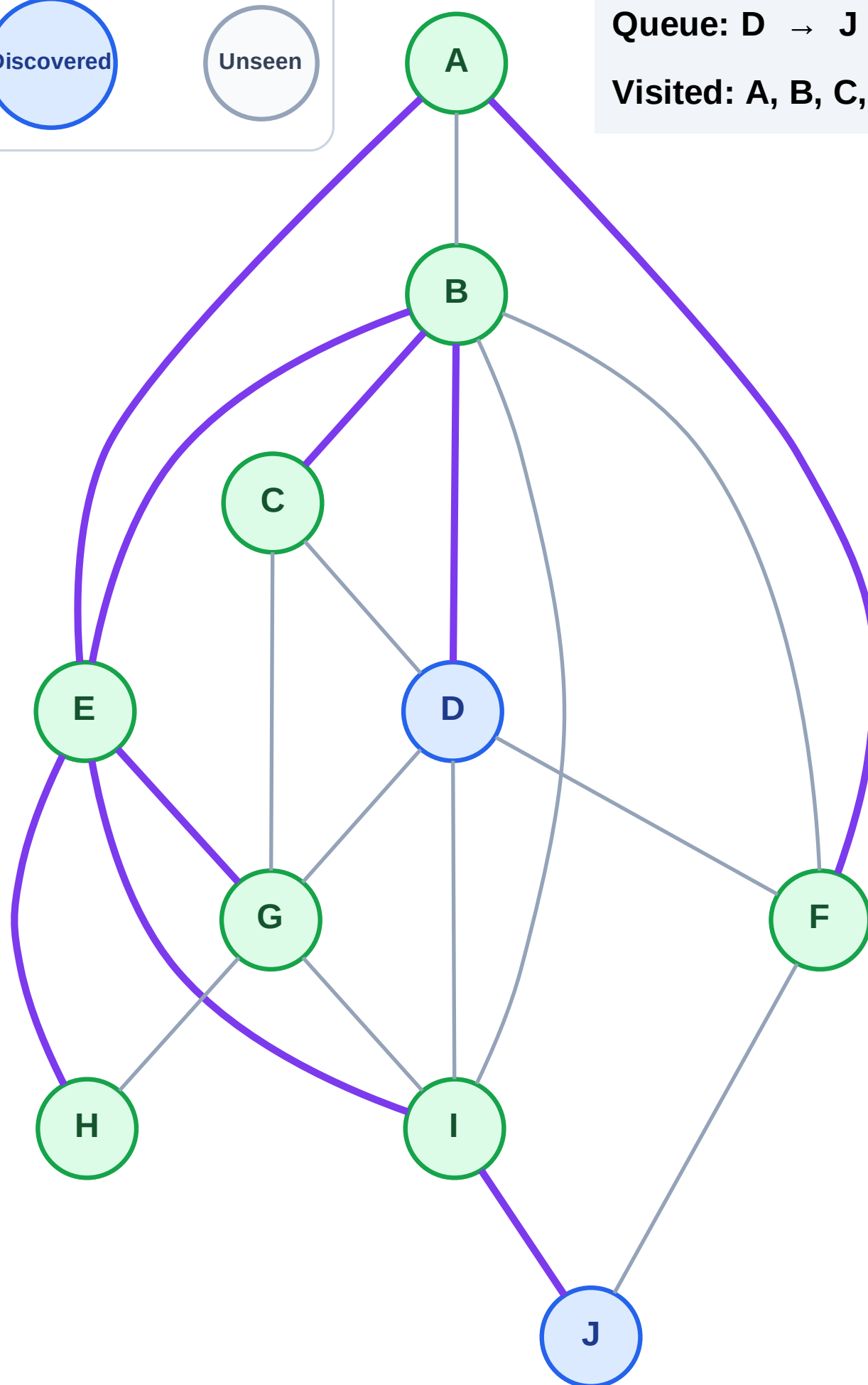
Step 17: No new neighbors from C

Legend



Queue: D → J

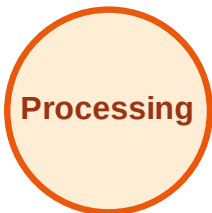
Visited: A, B, C, E, F, G, H, I



BFS Traversal

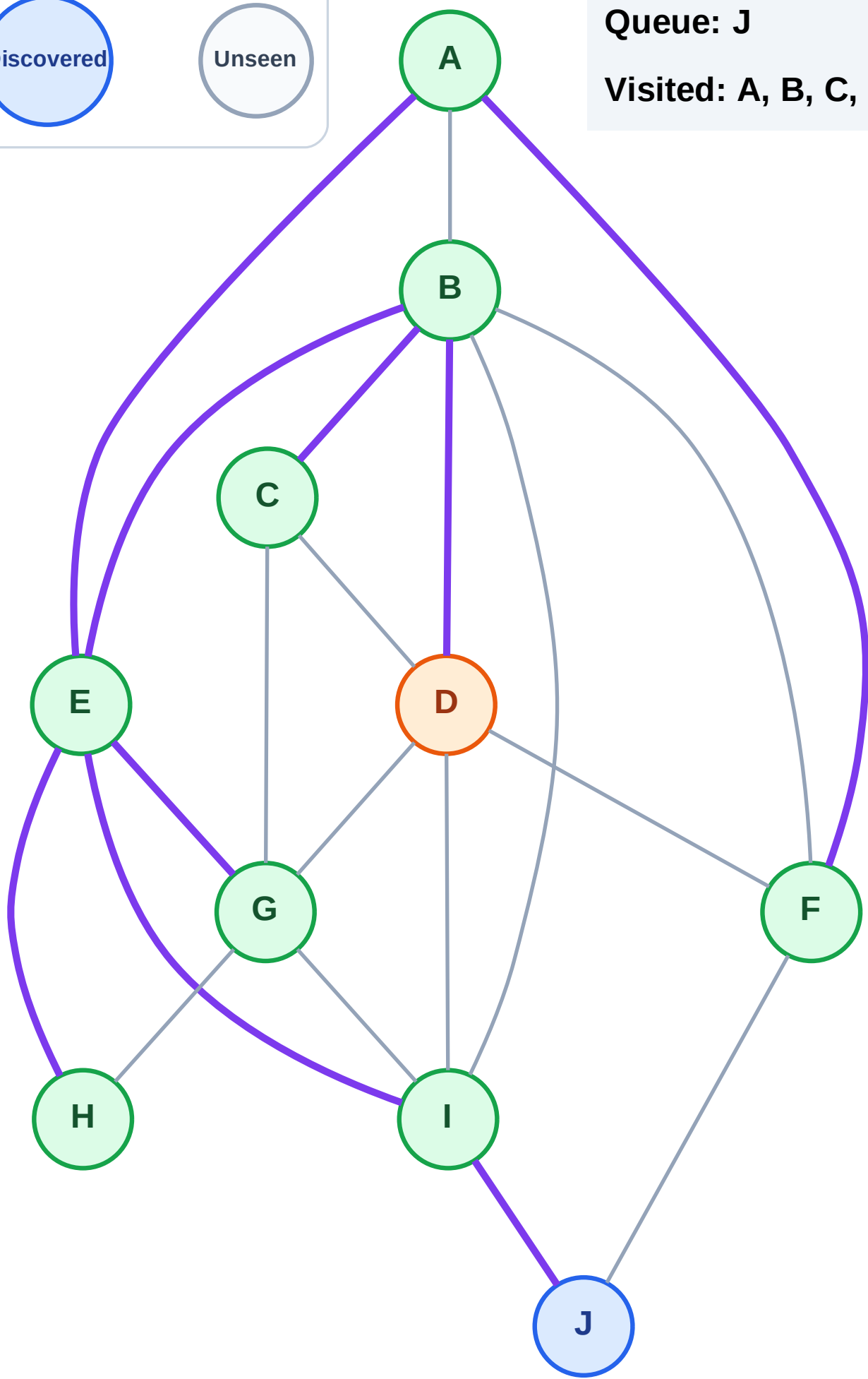
Step 18: Process node D

Legend



Queue: J

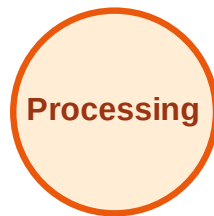
Visited: A, B, C, E, F, G, H, I



BFS Traversal

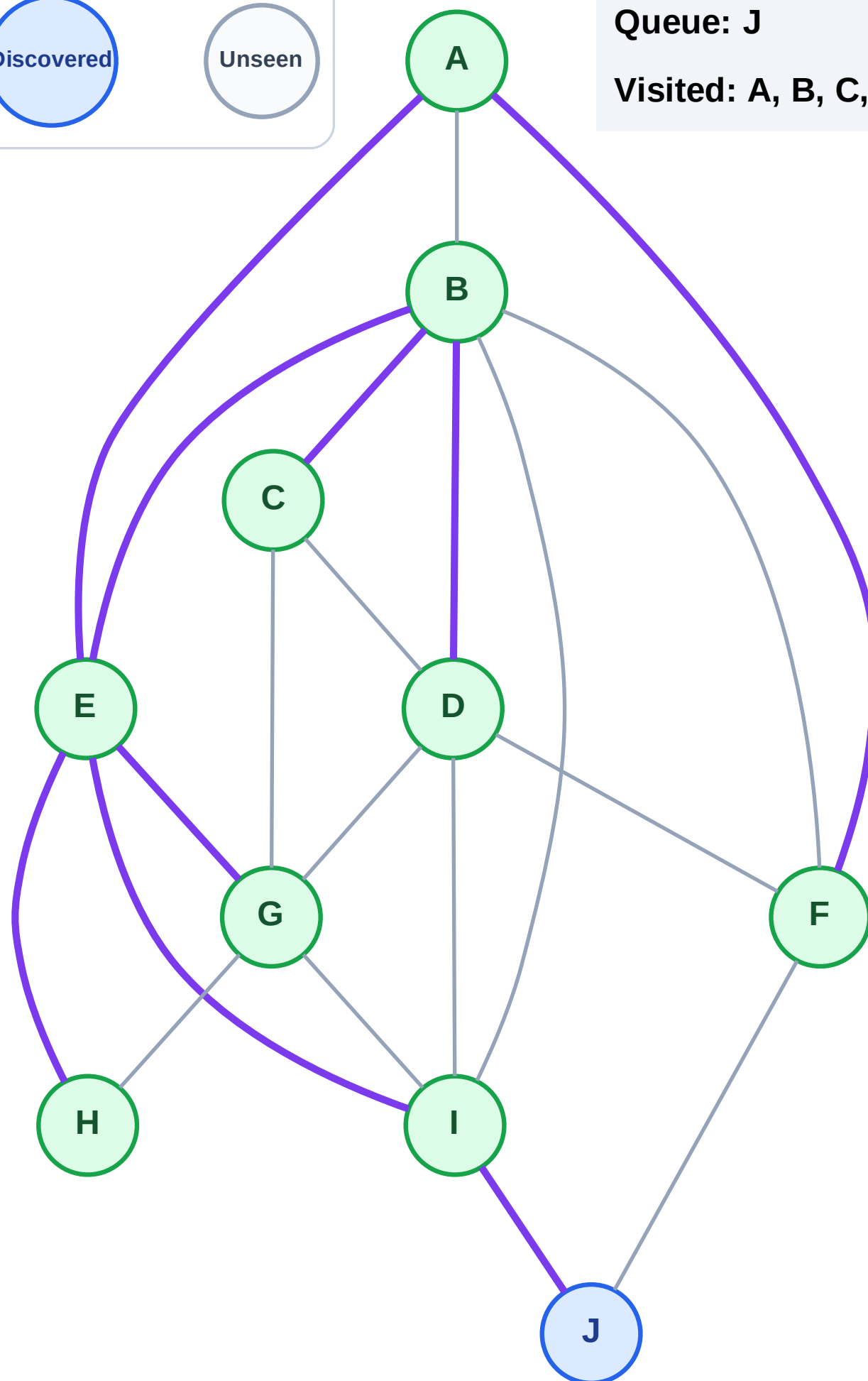
Step 19: No new neighbors from D

Legend



Queue: J

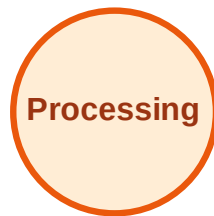
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

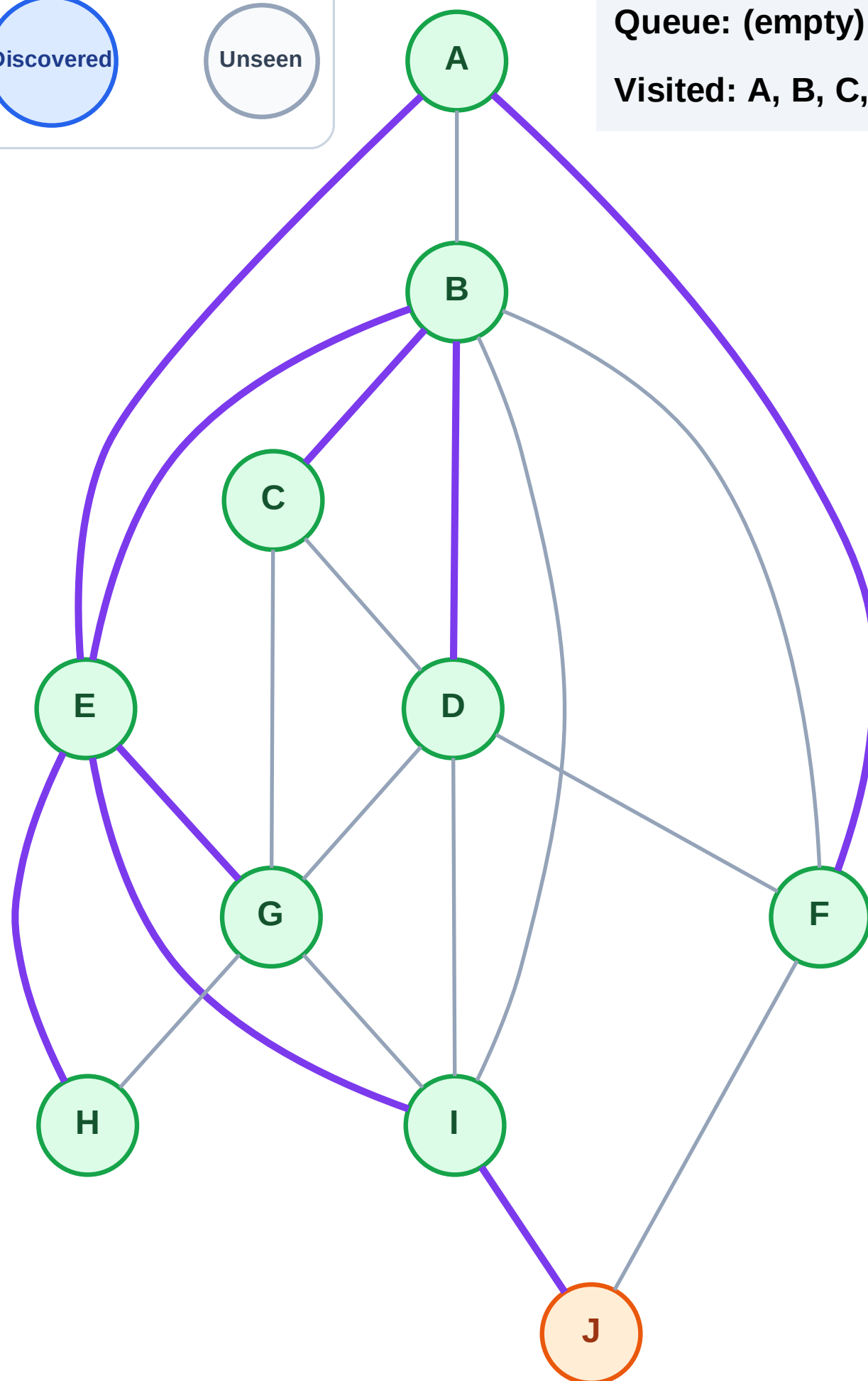
Step 20: Process node J

Legend



Queue: (empty)

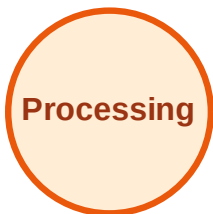
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

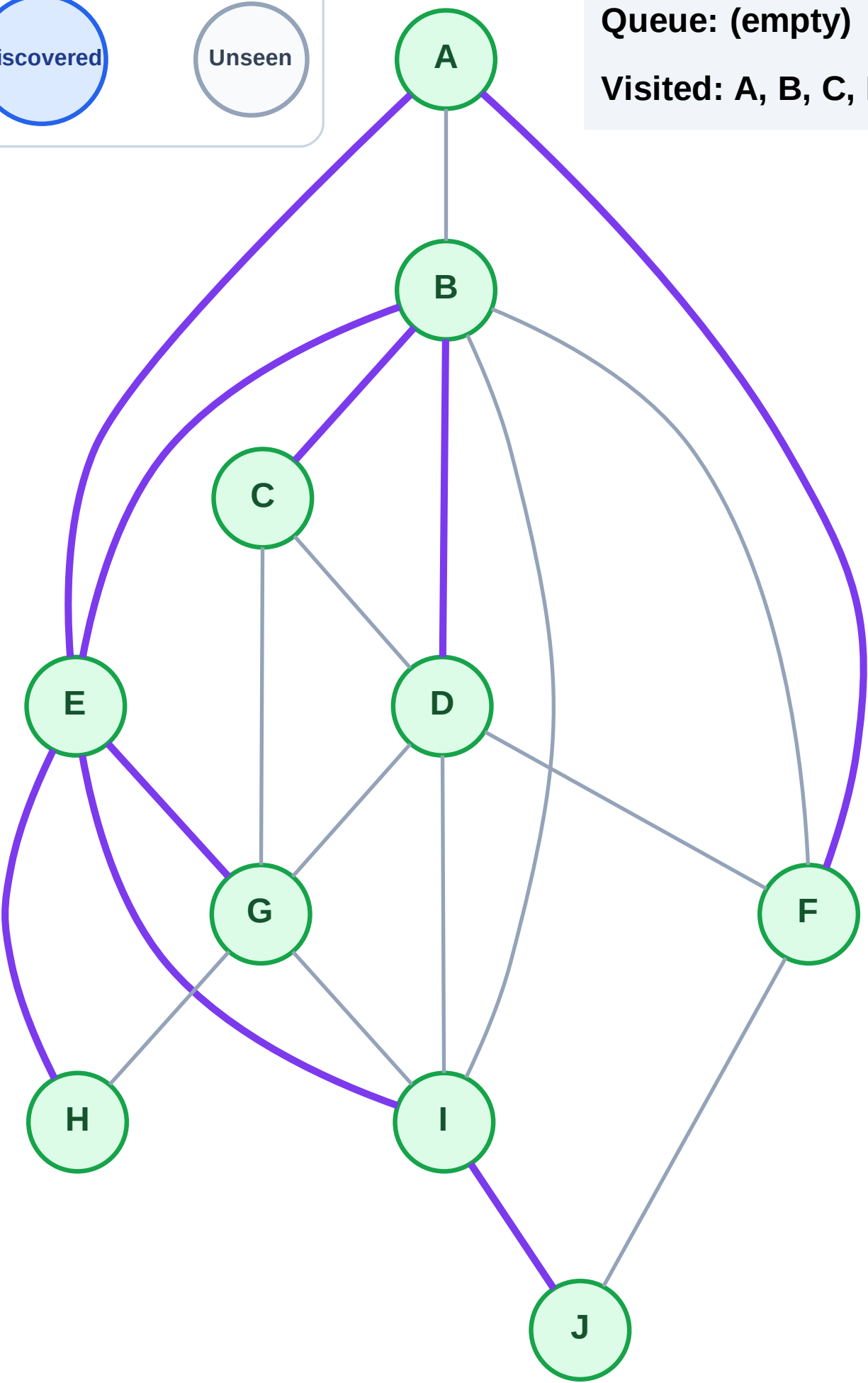
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

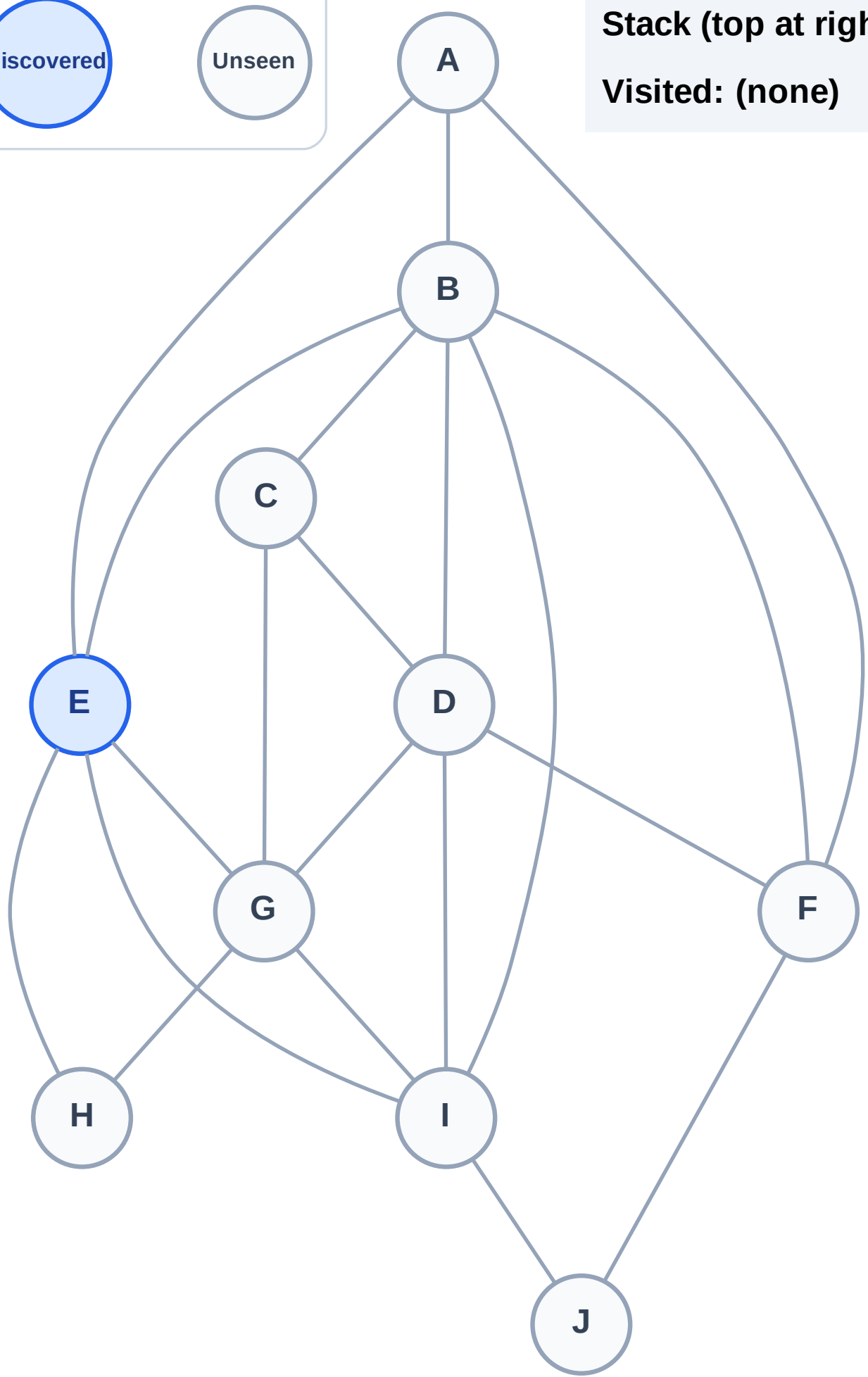
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

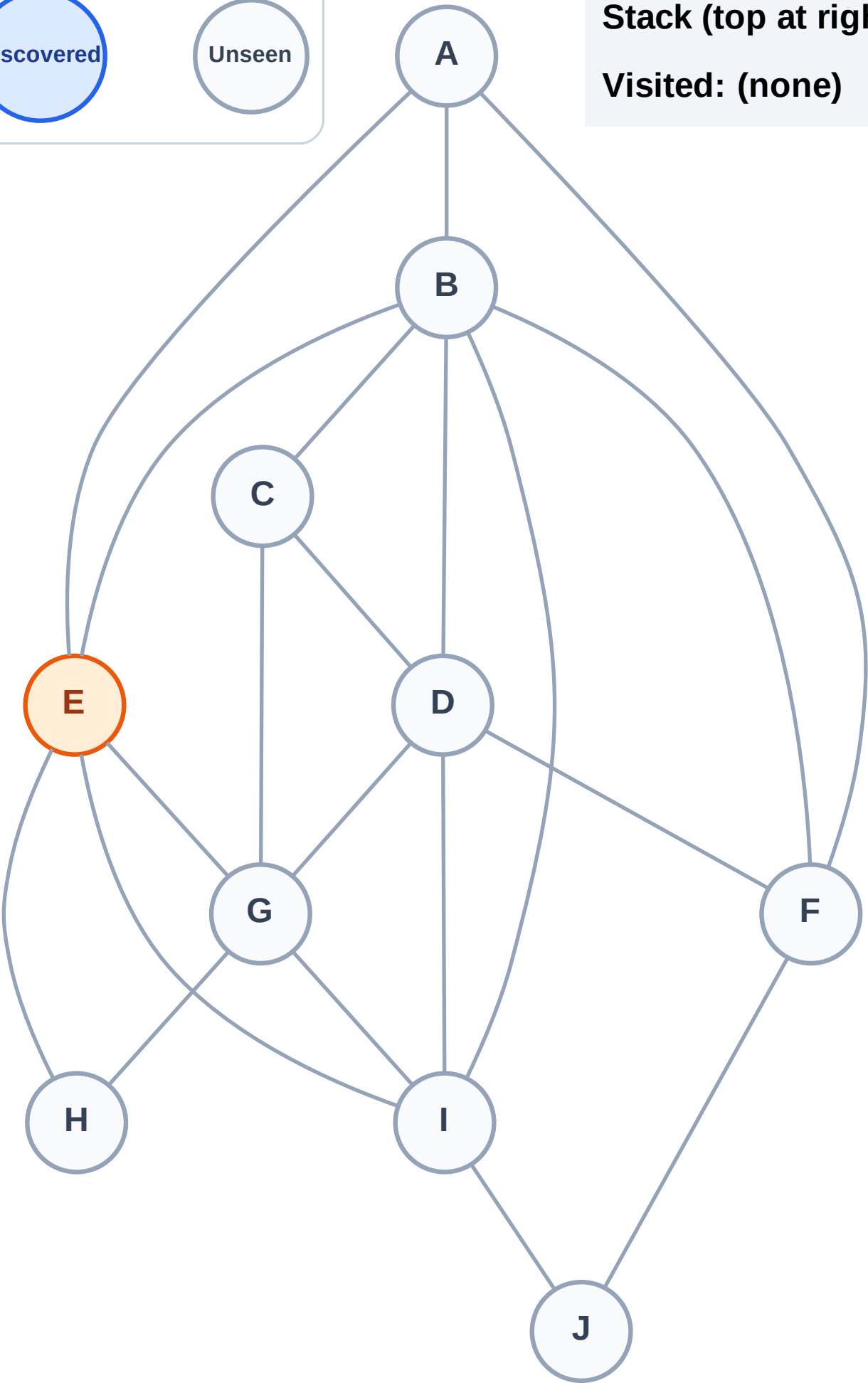
Complete

Processing

Discovered

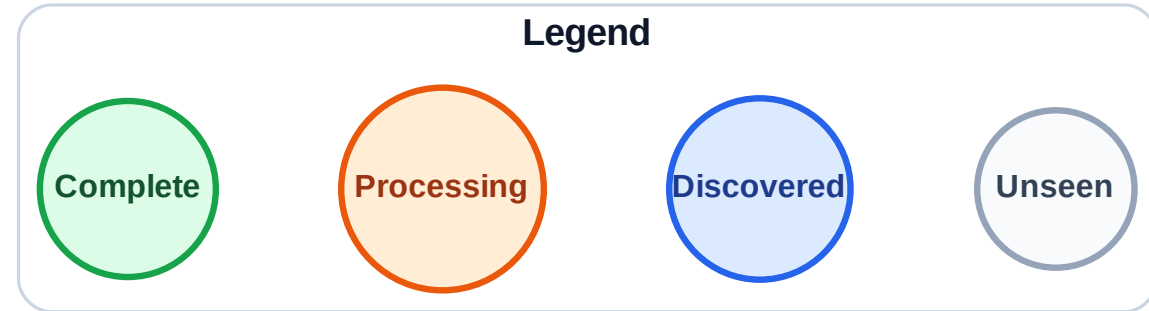
Unseen

Stack (top at right): (empty)
Visited: (none)



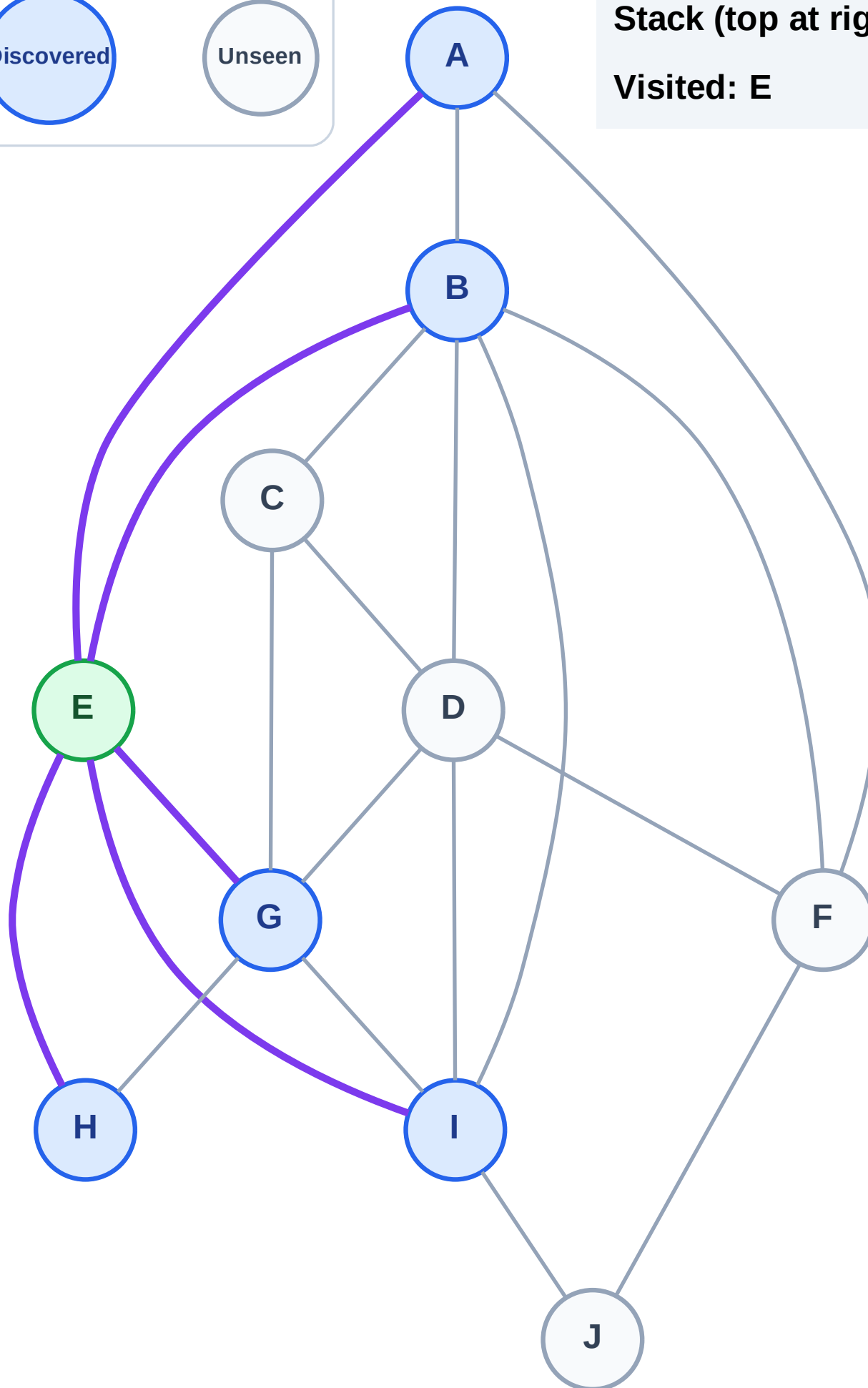
DFS Traversal

Step 3: Discover I, H, G, B, A from E



Stack (top at right): I | H | G | B | A

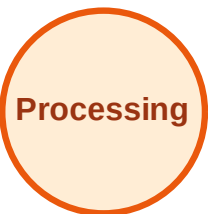
Visited: E



DFS Traversal

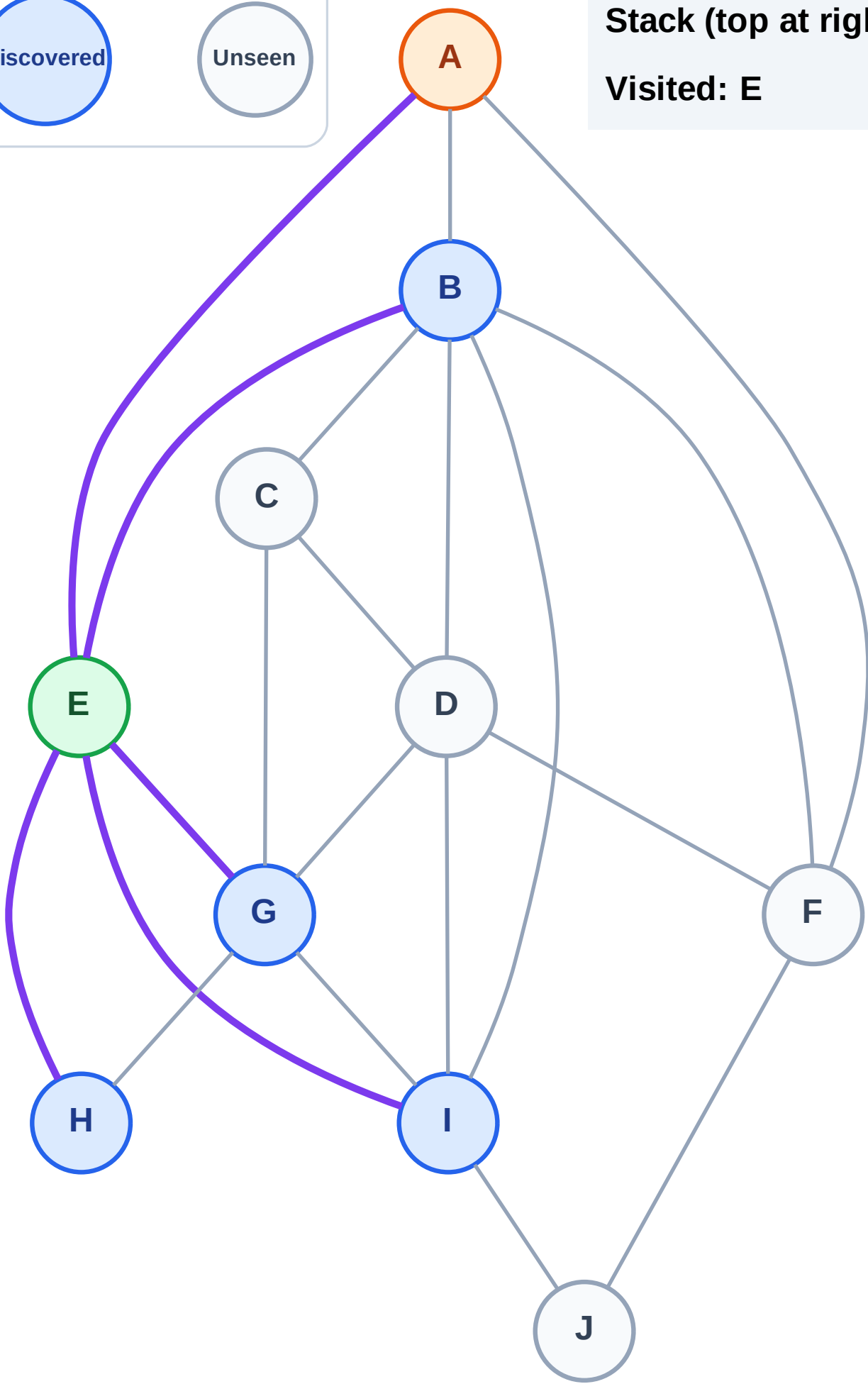
Step 4: Process node A

Legend



Stack (top at right): I | H | G | B

Visited: E



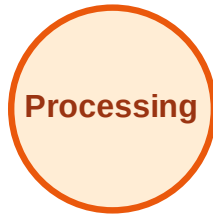
DFS Traversal

Step 5: Discover F from A

Legend



Complete



Processing



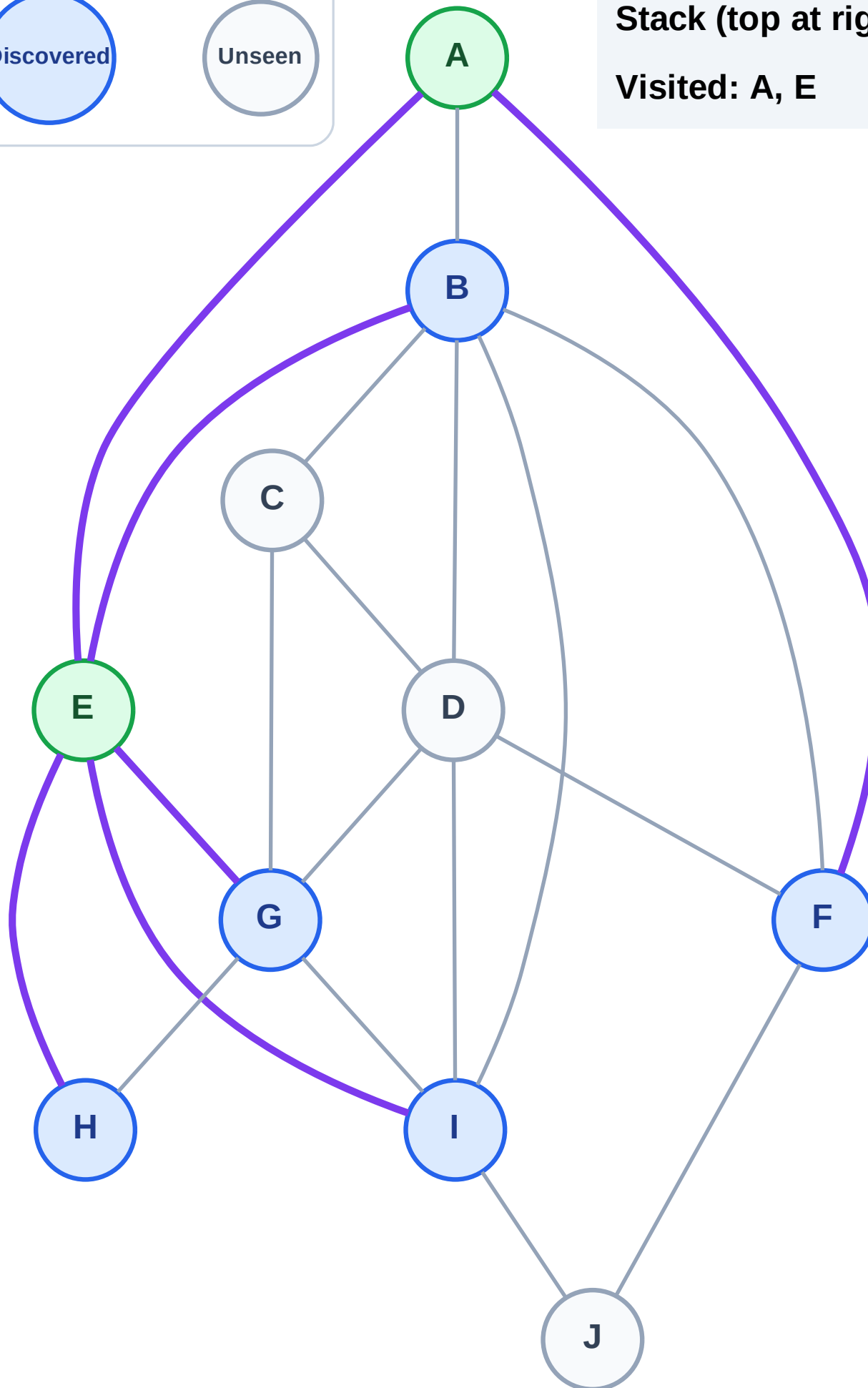
Discovered



Unseen

Stack (top at right): I | H | G | B | F

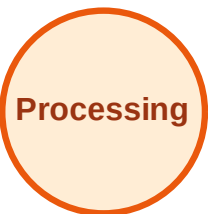
Visited: A, E



DFS Traversal

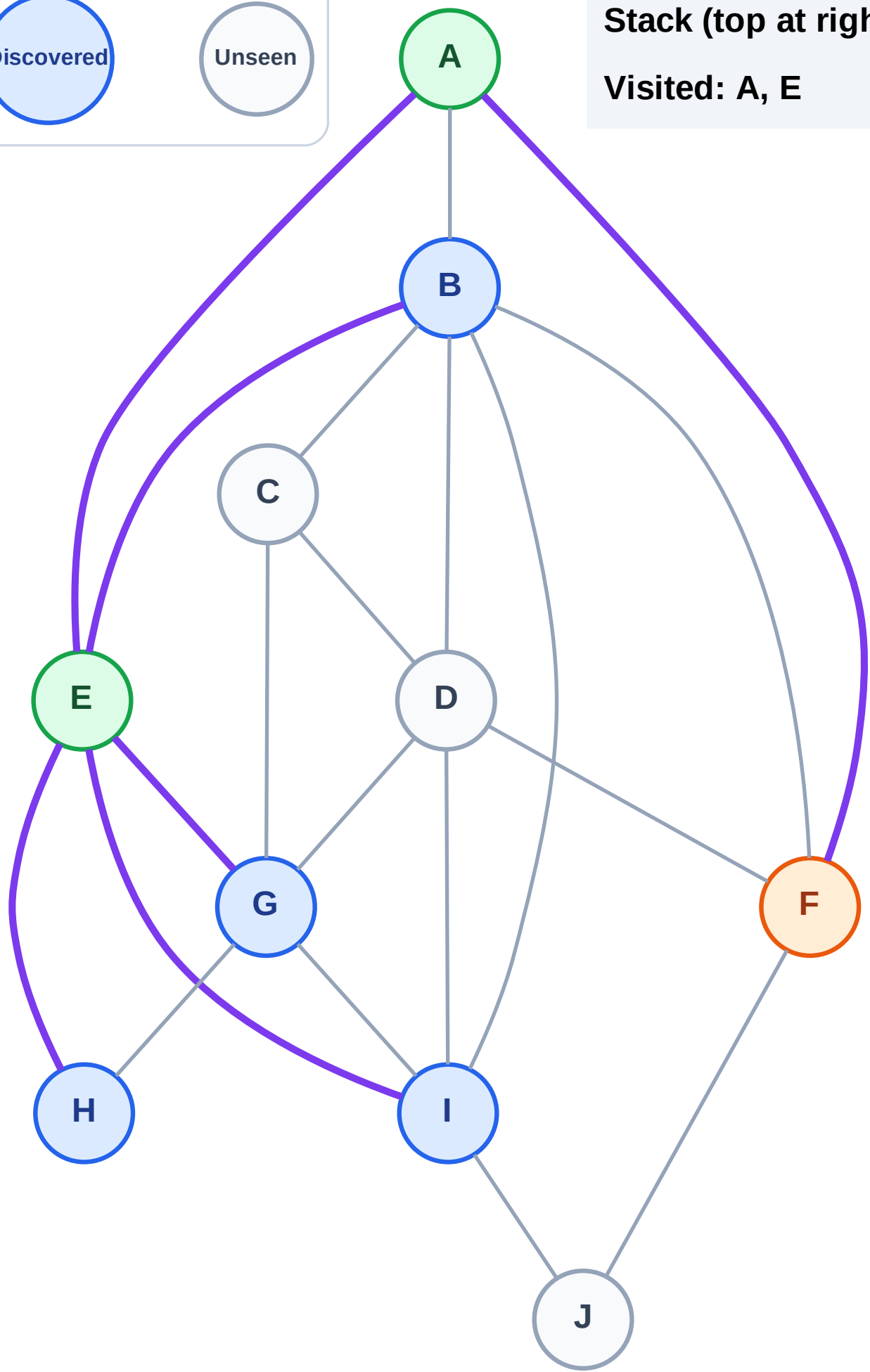
Step 6: Process node F

Legend



Stack (top at right): I | H | G | B

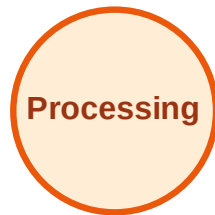
Visited: A, E



DFS Traversal

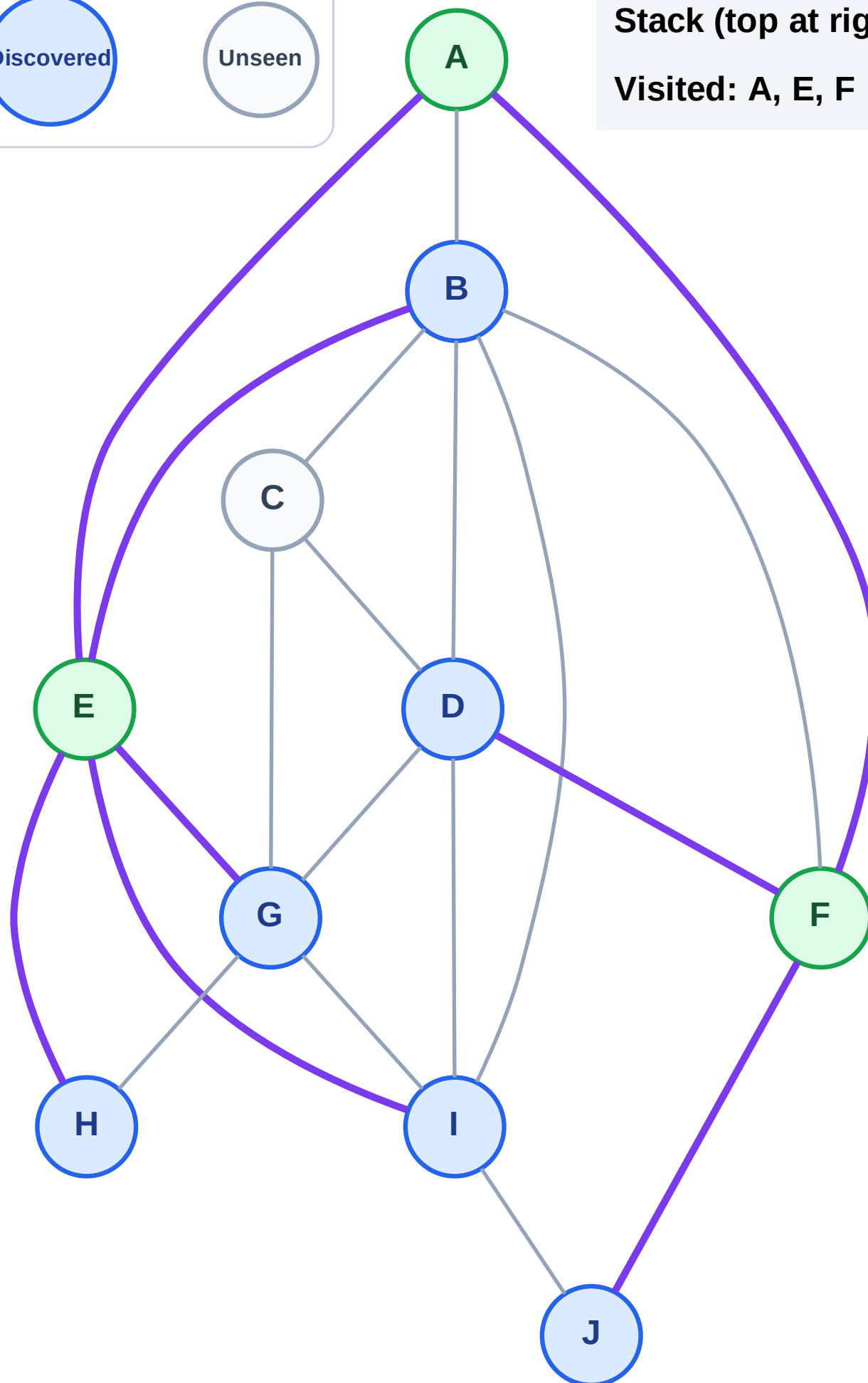
Step 7: Discover J, D from F

Legend



Stack (top at right): I | H | G | B | J | D

Visited: A, E, F



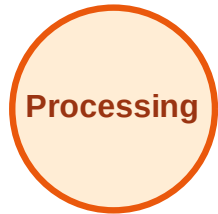
DFS Traversal

Step 8: Process node D

Legend



Complete



Processing



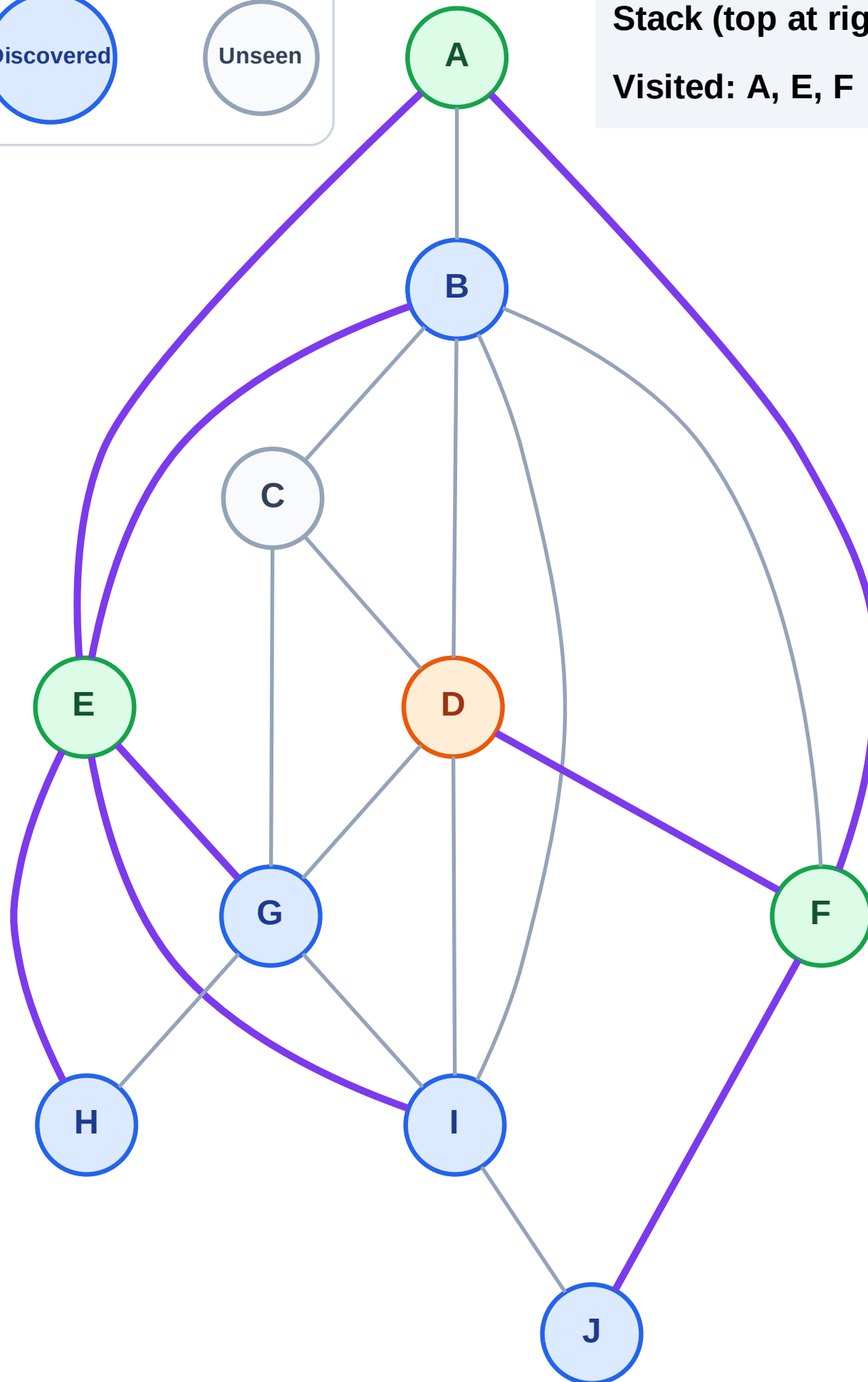
Discovered



Unseen

Stack (top at right): I | H | G | B | J

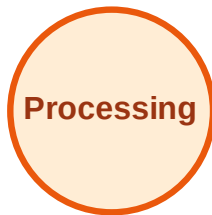
Visited: A, E, F



DFS Traversal

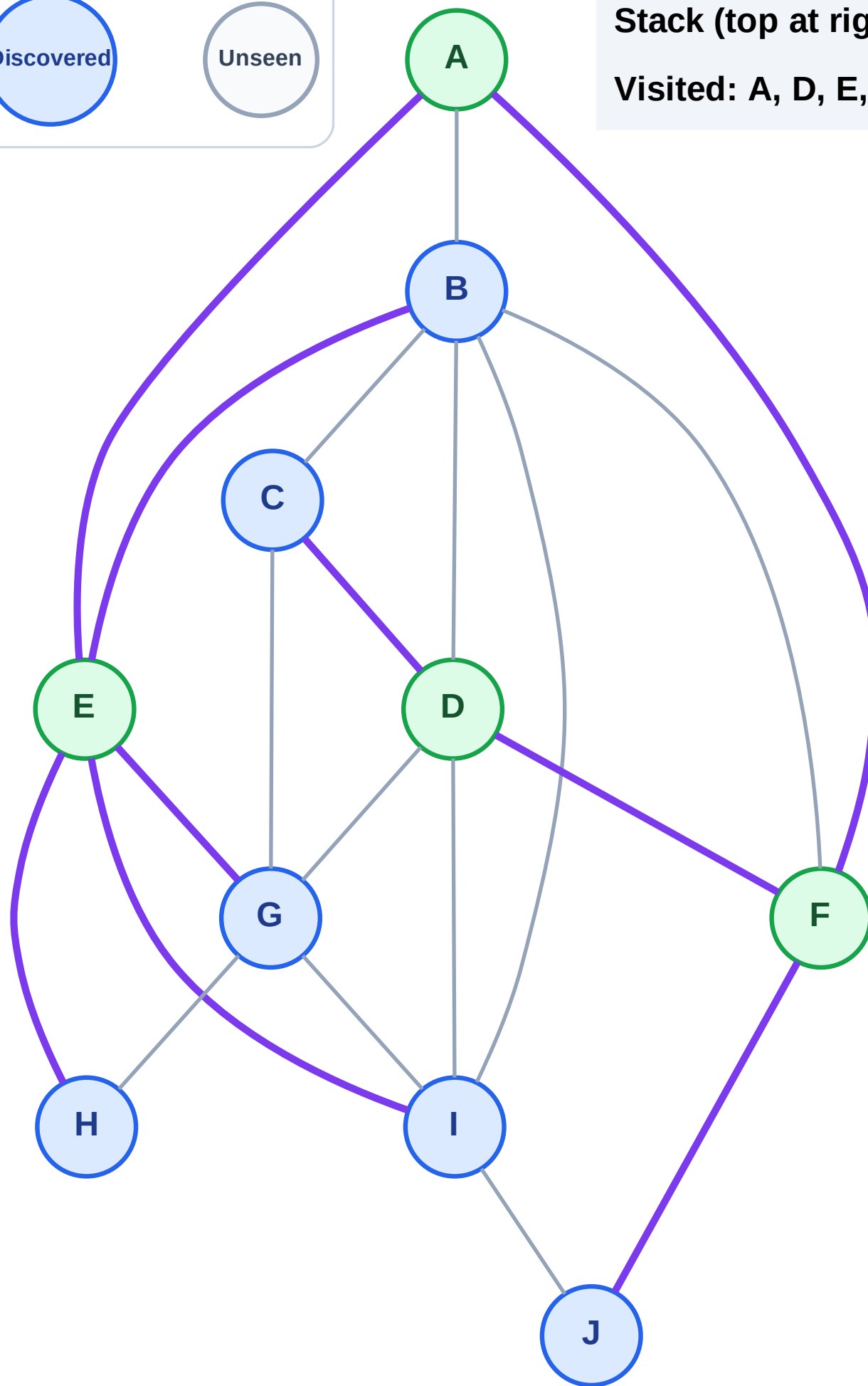
Step 9: Discover C from D

Legend



Stack (top at right): I | H | G | B | J | C

Visited: A, D, E, F



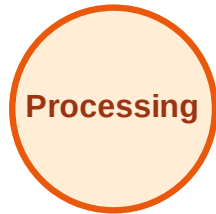
DFS Traversal

Step 10: Process node C

Legend



Complete



Processing



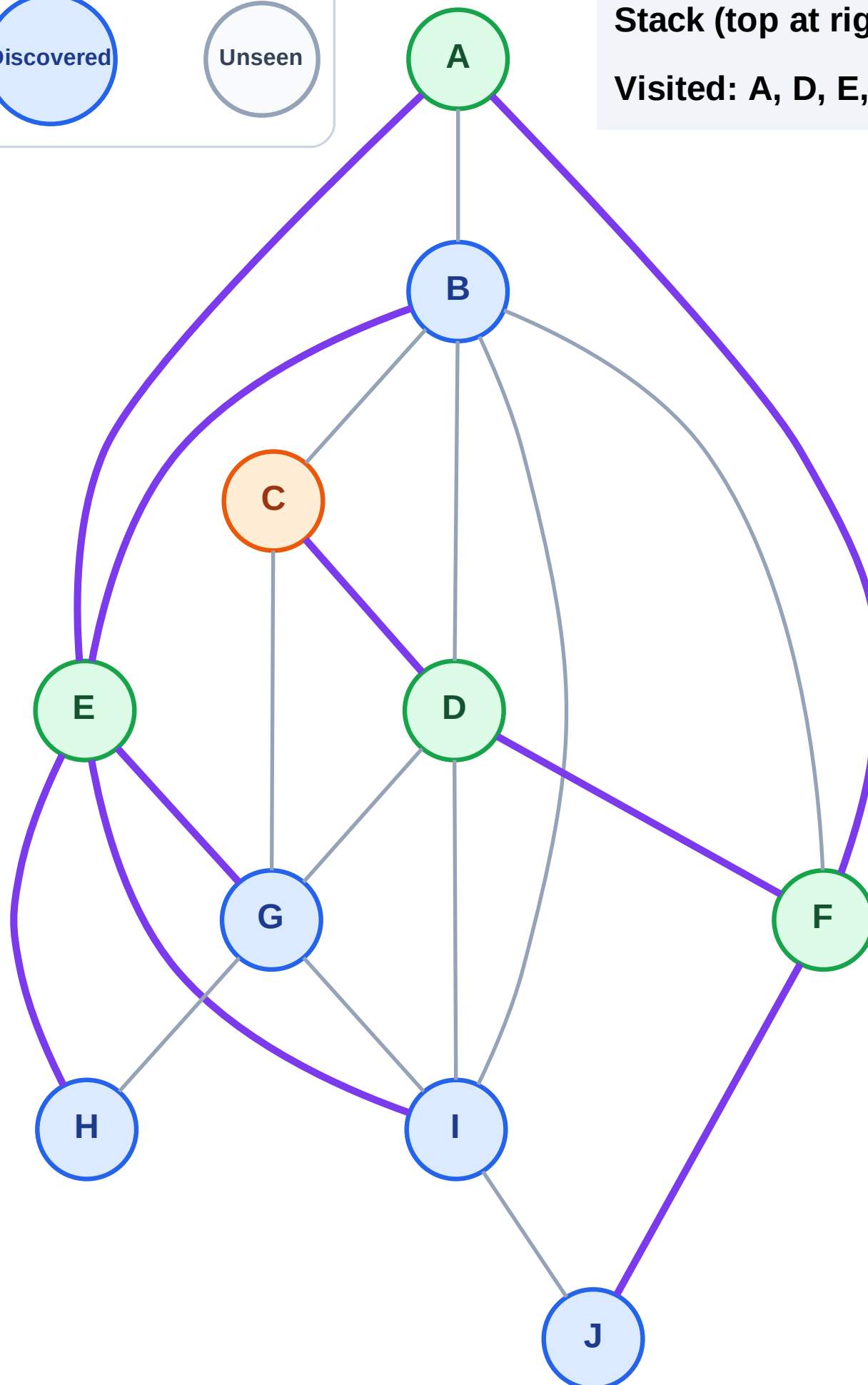
Discovered



Unseen

Stack (top at right): I | H | G | B | J

Visited: A, D, E, F



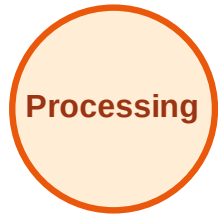
DFS Traversal

Step 11: No new neighbors from C

Legend



Complete



Processing



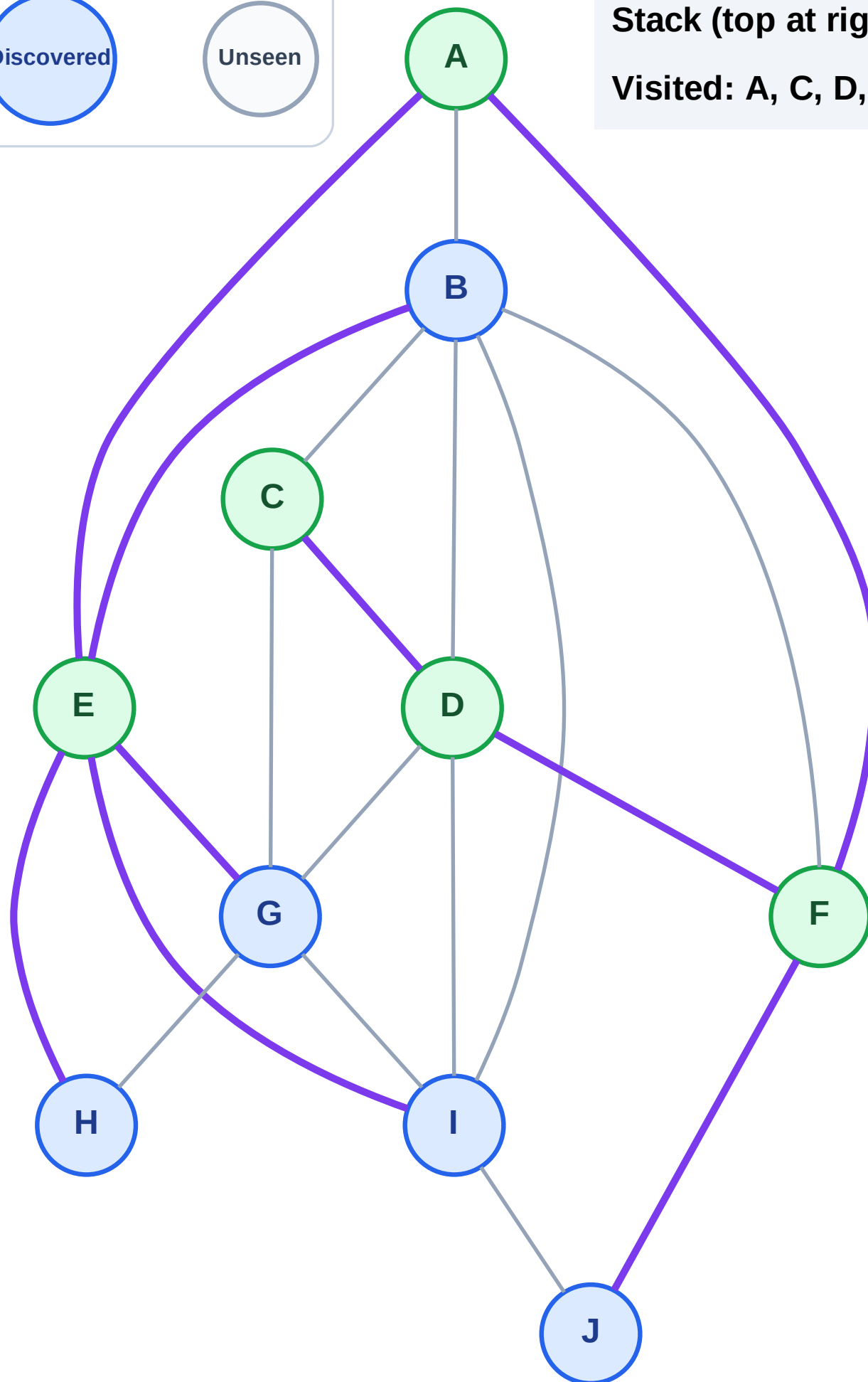
Discovered



Unseen

Stack (top at right): I | H | G | B | J

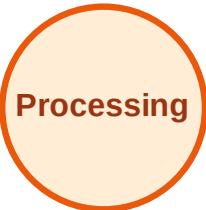
Visited: A, C, D, E, F



DFS Traversal

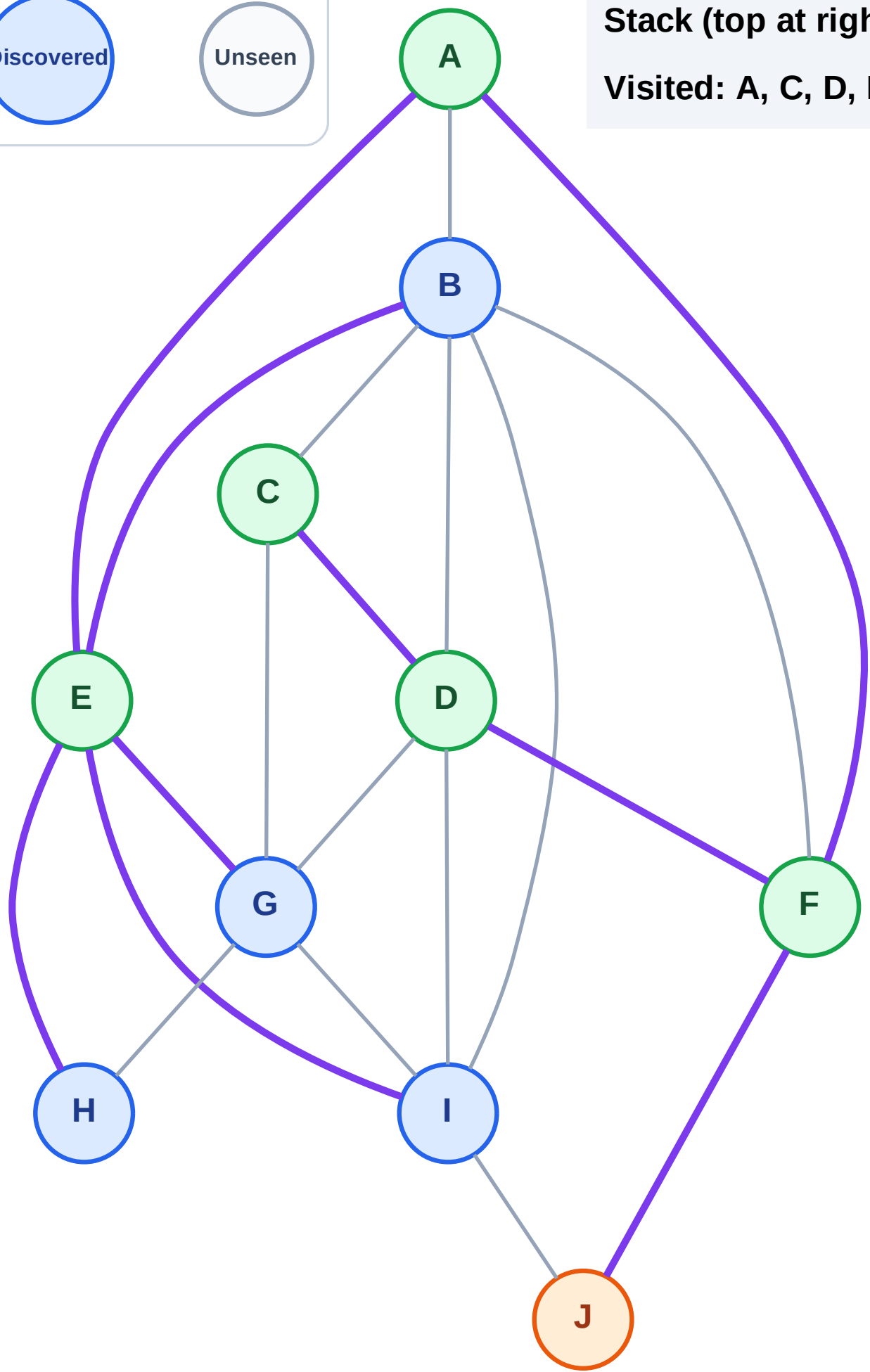
Step 12: Process node J

Legend



Stack (top at right): I | H | G | B

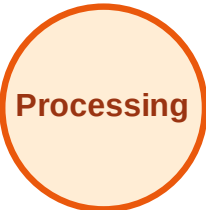
Visited: A, C, D, E, F



DFS Traversal

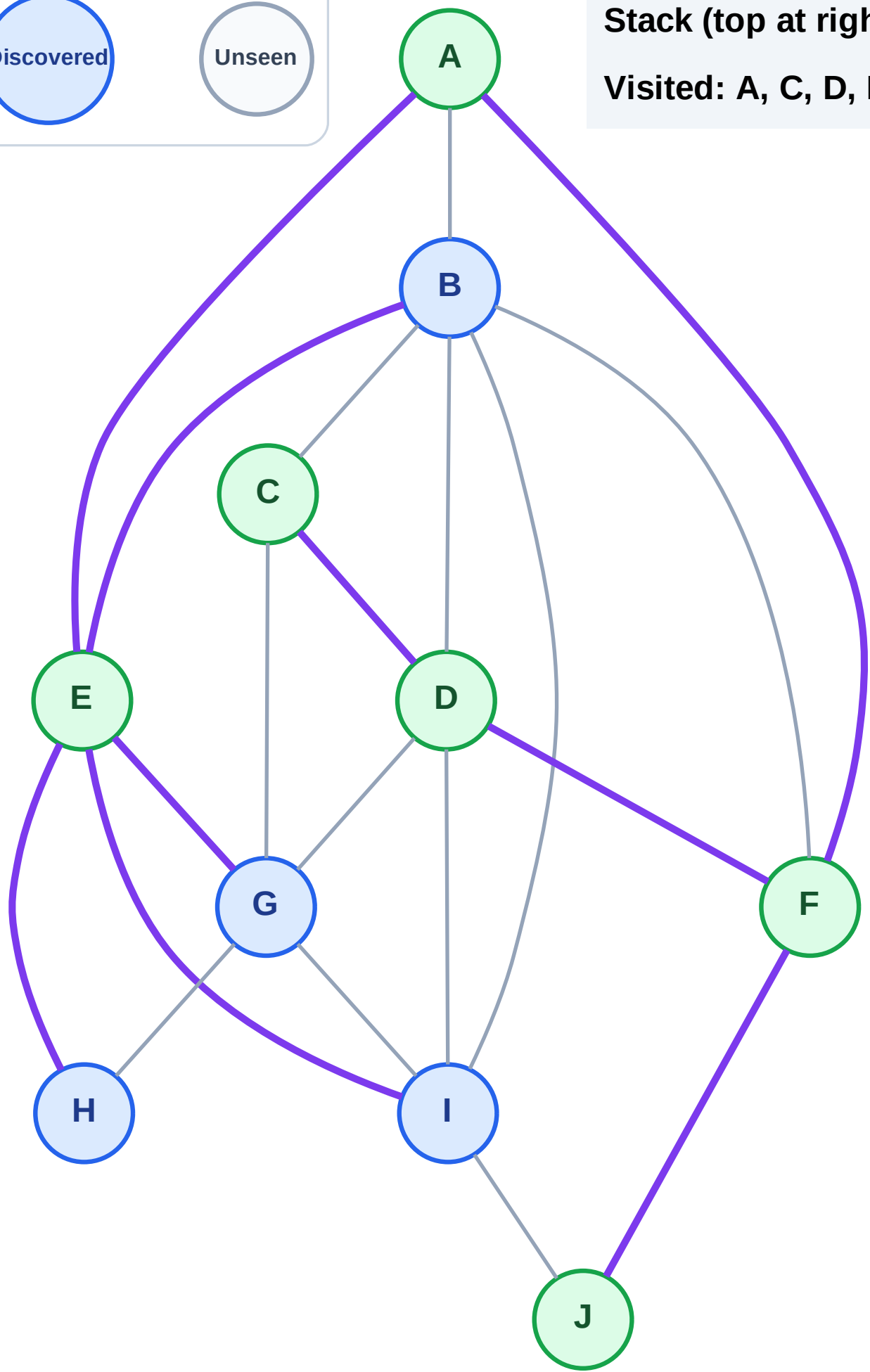
Step 13: No new neighbors from J

Legend



Stack (top at right): I | H | G | B

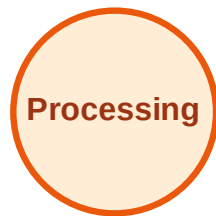
Visited: A, C, D, E, F, J



DFS Traversal

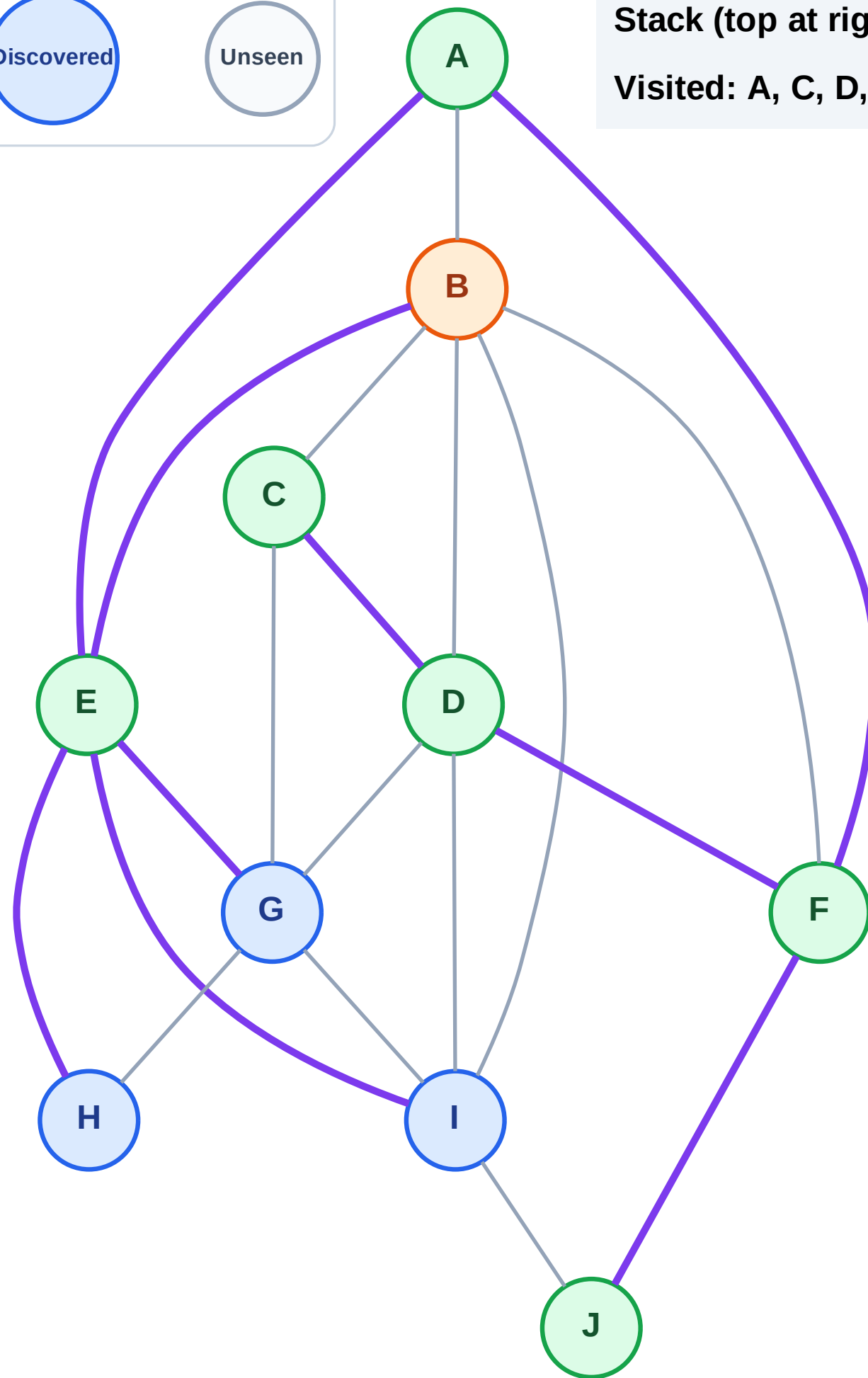
Step 14: Process node B

Legend



Stack (top at right): I | H | G

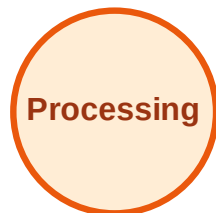
Visited: A, C, D, E, F, J



DFS Traversal

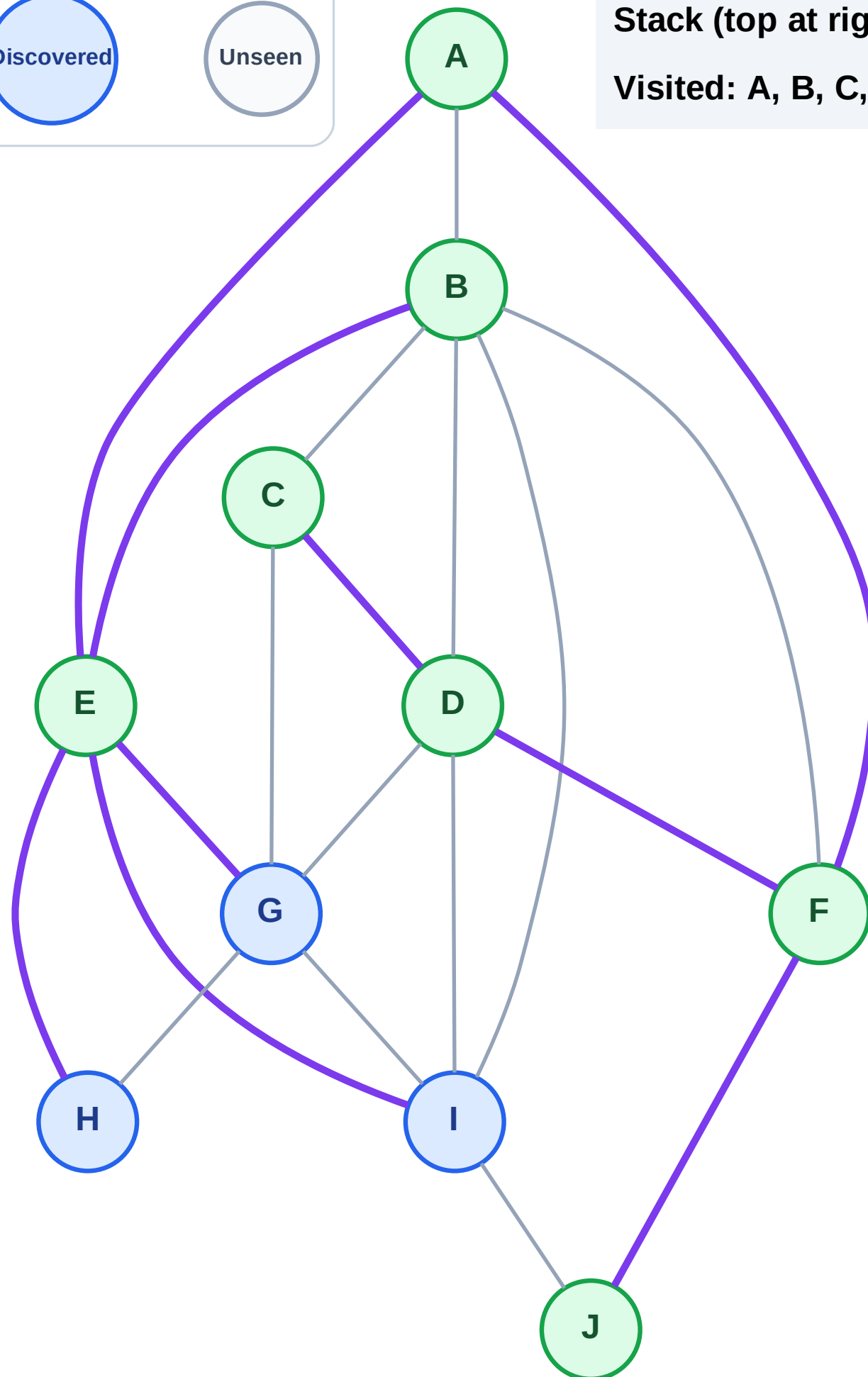
Step 15: No new neighbors from B

Legend



Stack (top at right): I | H | G

Visited: A, B, C, D, E, F, J



DFS Traversal

Step 16: Process node G

Legend

Complete

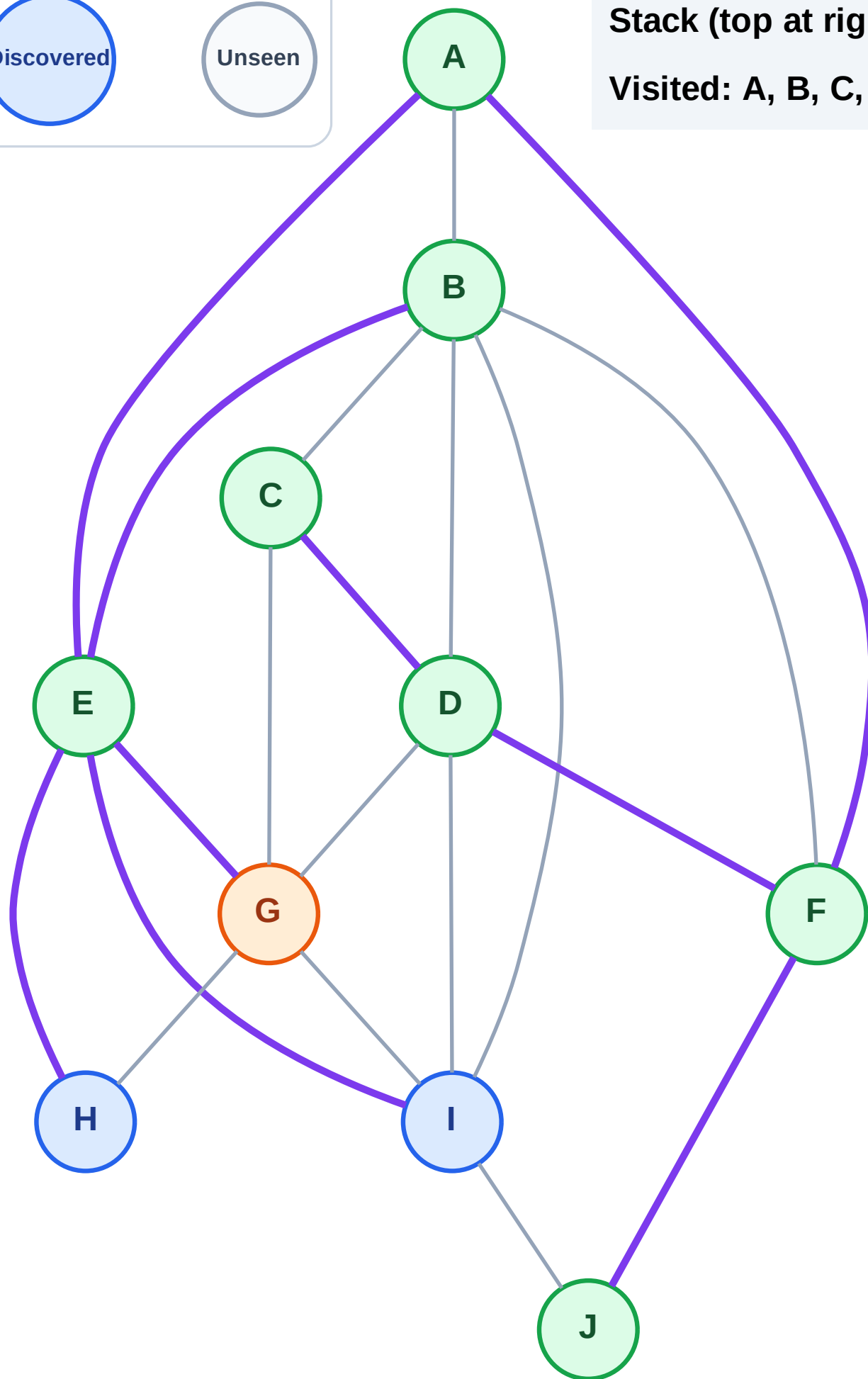
Processing

Discovered

Unseen

Stack (top at right): I | H

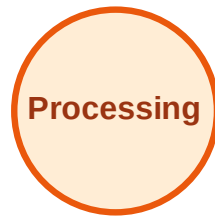
Visited: A, B, C, D, E, F, J



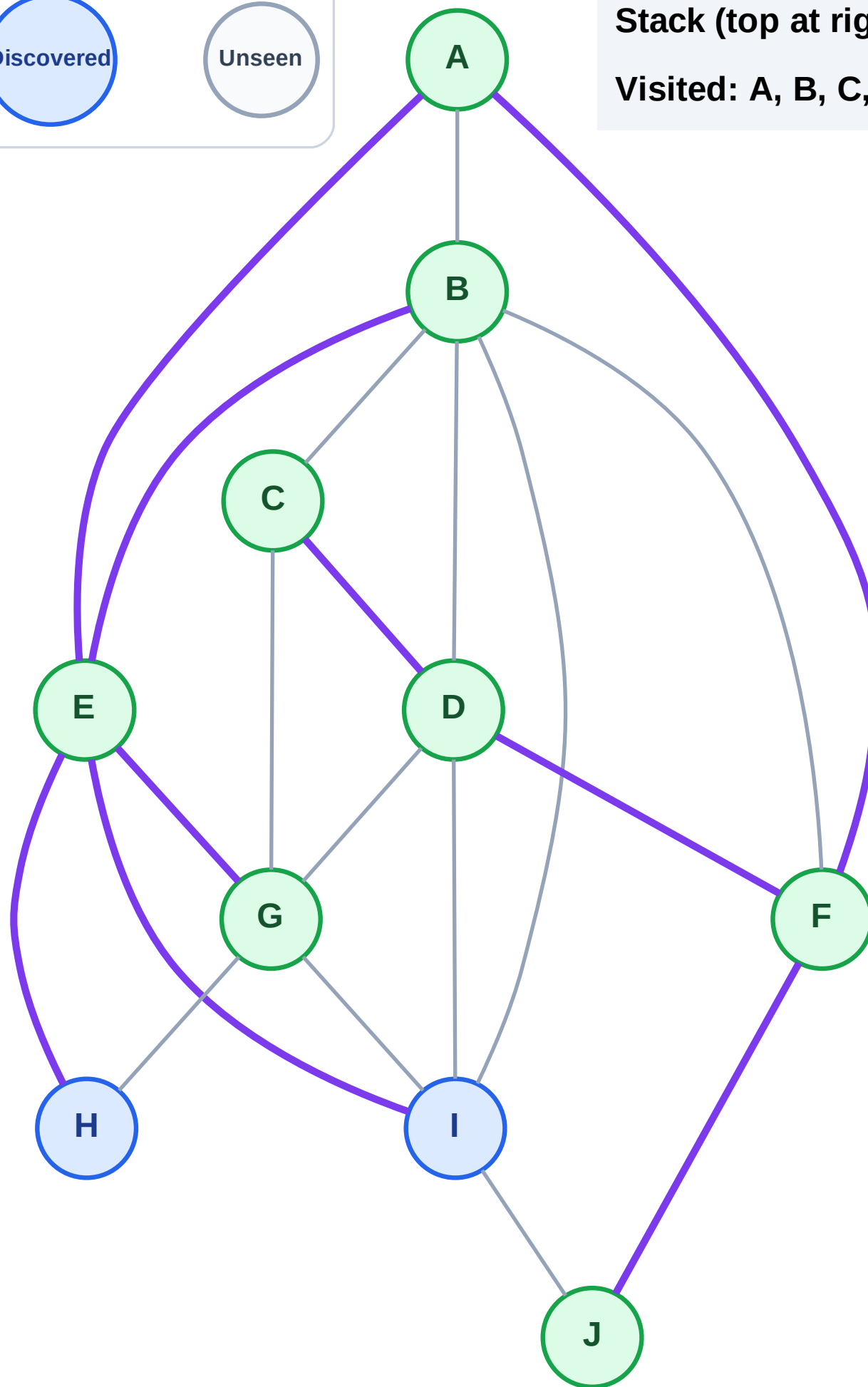
DFS Traversal

Step 17: No new neighbors from G

Legend

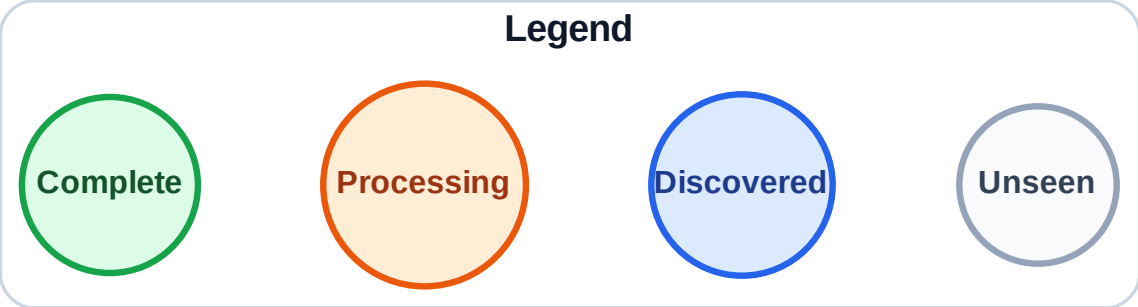


Stack (top at right): I | H
Visited: A, B, C, D, E, F, G, J

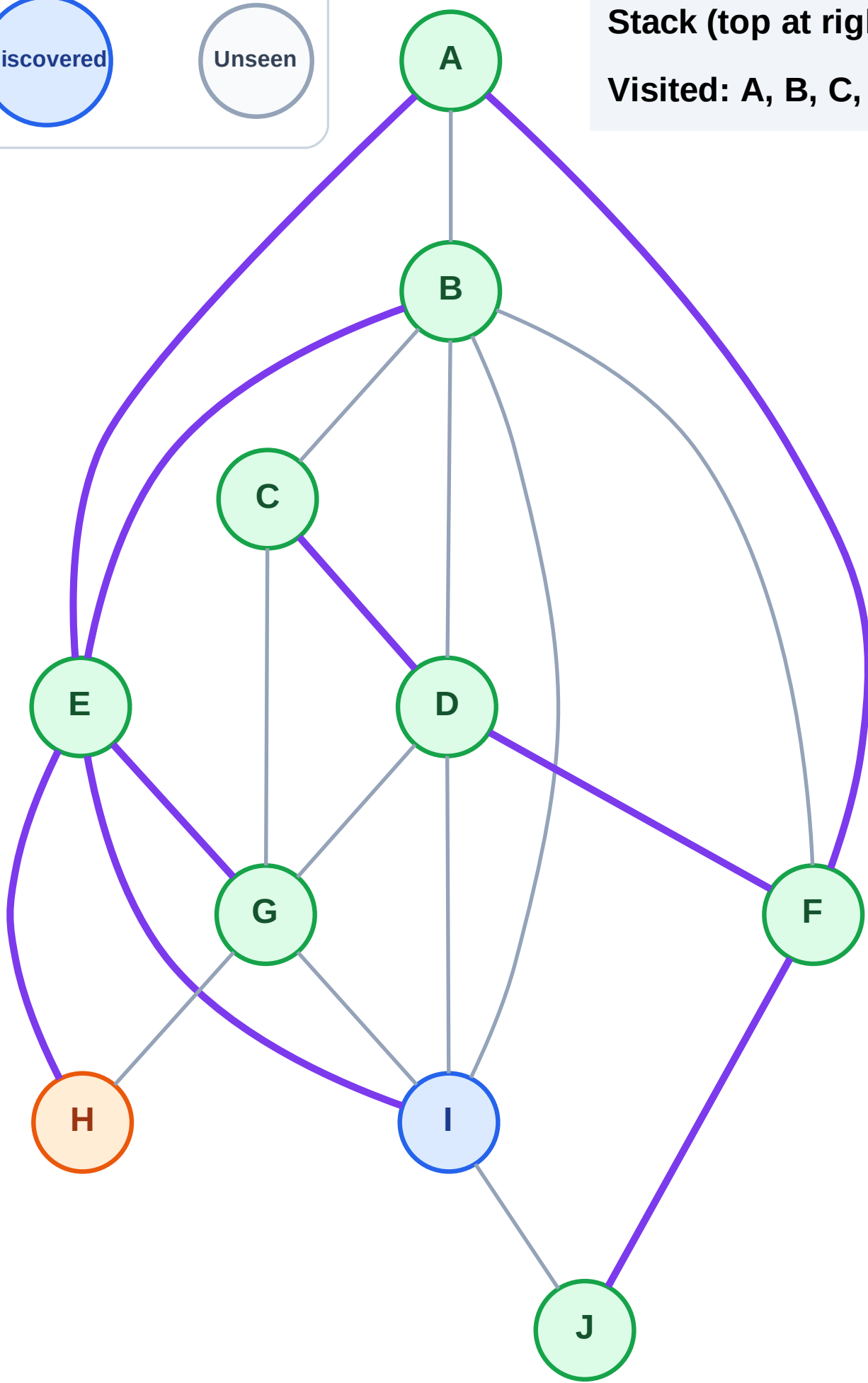


DFS Traversal

Step 18: Process node H



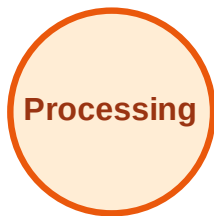
Stack (top at right): I
Visited: A, B, C, D, E, F, G, J



DFS Traversal

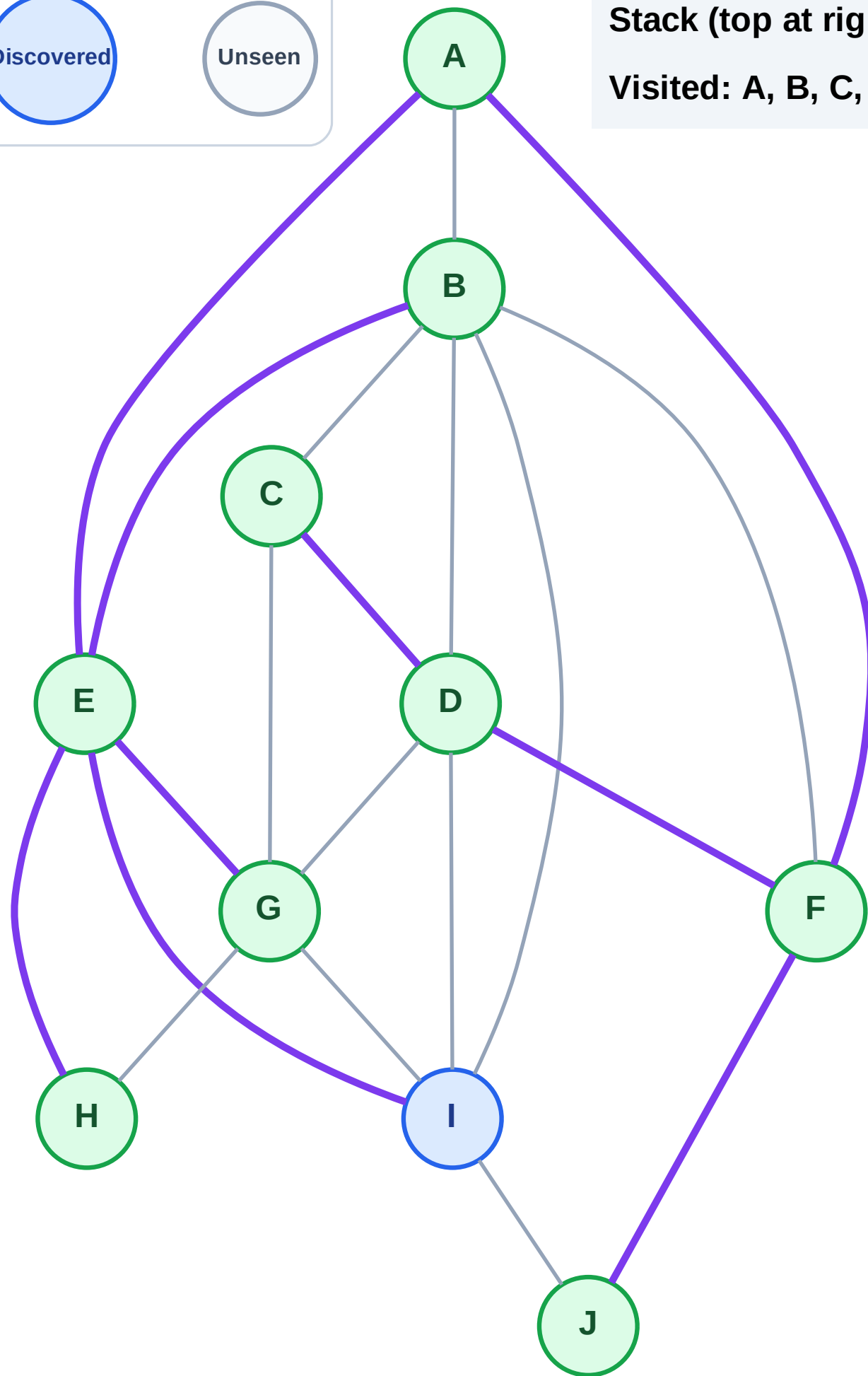
Step 19: No new neighbors from H

Legend



Stack (top at right): I

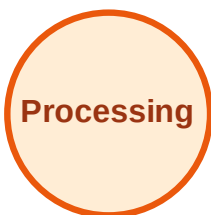
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

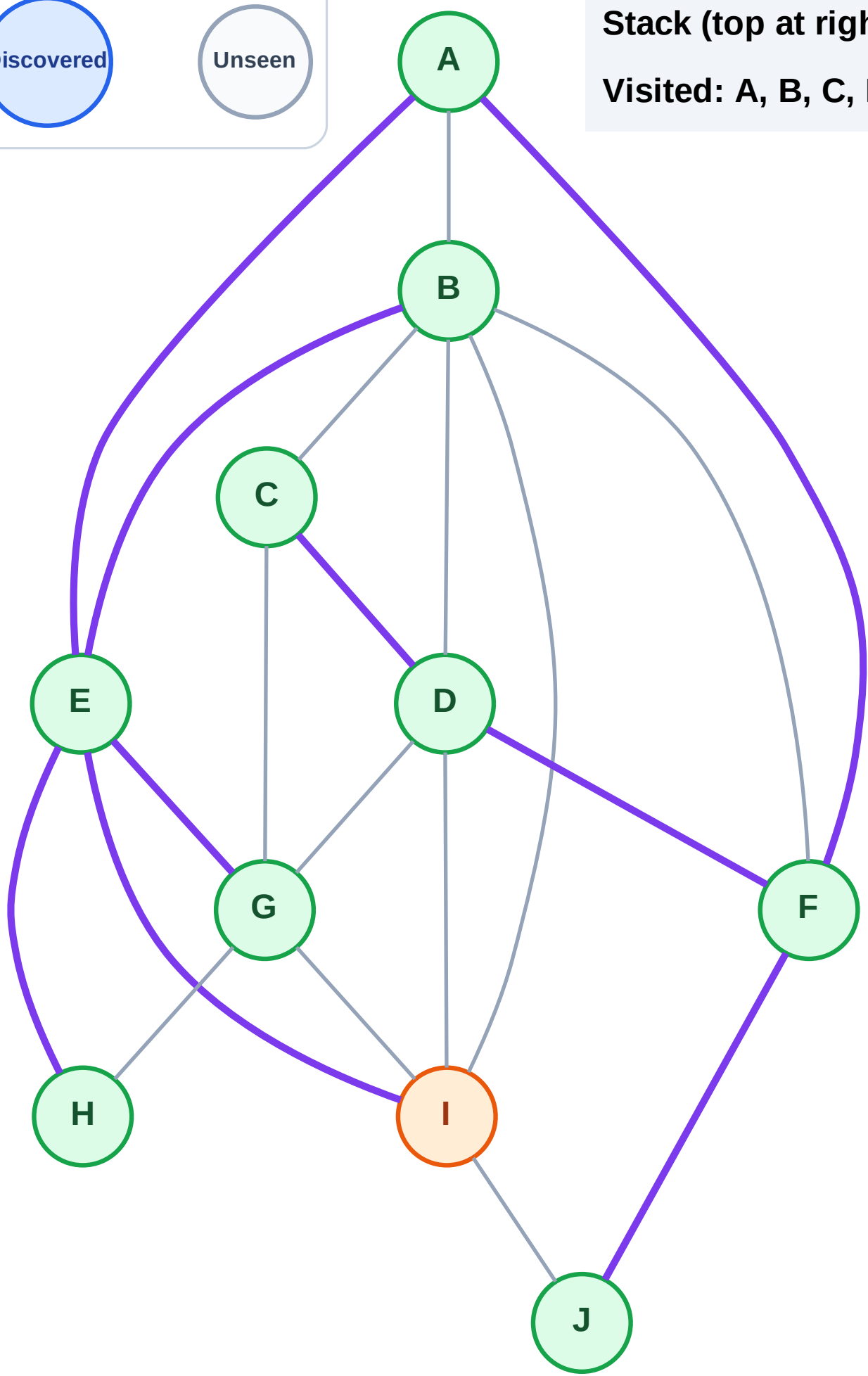
Step 20: Process node I

Legend



Stack (top at right): (empty)

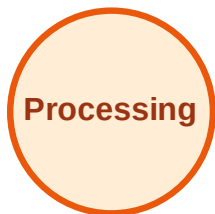
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

